A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft Trajectory

ISD 503 Capstone Project Report – Interim Report #2

Submitted to:
Mark Petrotta - Principal Engineer of Digital Engineering, MBSE, and Augmented Intelligence - Aerospace and Defense Systems
IOE Department of Integrated Systems Engineering & Design

Prepared by:
Owen O'Keefe
University of Michigan, ISD

October 11, 2026

Plan ref: to-do-list.md S1 — Confirm advisor/reader name and title for "Submitted to."

Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section efsec:impact), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report. State the two claims here as well as in the Introduction (Section efsec:introduction; owner's note on the 2026-10-11 print, page 11).

Contents

Executive Summary **1**

1 Introduction and Current State **3**
- 1.1 Overview of the Project . . . 4
- 1.2 Review of Current State Design / Literature Review . . . 5
- 1.3 Stakeholders . . . 7
- 1.4 Scope . . . 10
- 1.5 Project Deliverables . . . 13
- 1.6 Project Timing . . . 13

2 Assumptions and Methodology **13**
- 2.1 Systems Engineering Approach . . . 13
- 2.2 Working Assumptions . . . 16
- 2.3 Proposed Airline-to-ATC Interface . . . 17

3 Experiment **17**
- 3.1 Scenario . . . 17
- 3.2 Cases and Resolution Options . . . 19
- 3.3 Scoring . . . 20
- 3.4 Aircraft Model and Its Simplifications . . . 21
- 3.5 Worked Example: Option O4, the Light Aircraft Climbs 2,000 ft . . . 23

4 Results and Discussion (Preliminary) **24**
- 4.1 Operational Concept . . . 24
- 4.2 Stakeholder Objectives . . . 26
- 4.3 Architecture: Detailed and Abstracted . . . 26
- 4.4 Experiment Results . . . 30
- 4.5 Trace Back to the Model . . . 31
- 4.6 Emissions and Who Pays . . . 31

5 Impact/Financial Benefits **32**

6 Conclusions and Recommendations **32**

A Appendices **40**
- A.1 Full Nominal-Flight ConOps . . . 40
- A.2 Full Stakeholder Register and RACCI Matrix . . . 40
- A.3 Full SysML Diagram Set . . . 40

A.4 Full Myopic-Optimization Conflict Catalog 40
A.5 Optimization Experiment Detail 40
A.6 Generative AI Use Disclosure 40
A.7 Source Register 41

1 Introduction and Current State

Move to sub-section.

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Introduction. Interim Report #2 resubmits this section updated. Status after the 2026-10-11 review: (1) the mid-September "current state" paragraphs in Section 1.2 are replaced by an AI-drafted October version for the owner to rewrite; (2) the research-gap placeholder below still needs the owner's paragraph; (3) the project timing subsection should show the nine turns, not only the August table.

Plan ref: to-do-list.md Track C (D-016, D-017) — State the two claims here, briefly, and again in the Executive Summary (owner's note on the old "Two Claims" subsection: "Why make this its own section? Shouldn't these be embedded within the Exec Summary and Intro, probably duplicated?"). Claim 1, explored by the experiment: replacing the decider's low-fidelity rule of thumb with a higher-fidelity model fed by weight and cost index across the airline-to-ATC interface changes which aircraft is moved and lowers total system cost (whether one extra clearance is worth it is a result, not an assumption). Claim 2, shown by the model: the architecture locates that interface, names who owns the decision and the data, states the unmet requirement, and lets the scoring model's fidelity be swapped (conceptual, nominal mass, actual mass) with the structure unchanged. The simulation would give the same numbers without the model; say so, and say what the model contributes that the numbers do not.

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

This project builds a reference MBSE architecture for how airspace intent is managed across the National Airspace System (NAS). The architecture spans the organizations that define and enforce rules, the domains that manage airspace resources, the systems that support flight operations, and the onboard systems that turn intent into an executable trajectory and guidance behavior.

DeLaurentis [16] states the problem as follows:

> "Problems of significant complexity are facing decision-makers within government and industry, and the moniker of 'system-of-systems' (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the 'stovepipe' mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems ... A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time."

The statement dates from 2005 and still describes the NAS today. The NAS is a system of systems, and the decisions its stakeholders make have consequences for the safety, efficiency, and sustainability of air transportation.

Airlines invest significant time and money in operational plans that maximize revenue. On the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it [11]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of specialists in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to limit the effect of a disruption on the airline's schedule and revenue.

The literature does not converge on one standard definition of the "cost" the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. The optimization

literature commonly models delay cost as a linear function of delay minutes, but Hu et al. [44] note that this is inconsistent with airline practice, where delay cost is a nonlinear function of delay time, and that, faced with this complexity, airlines tend to put the speed of a recovery decision ahead of the mathematical optimality of the solution. The same preference for fast, tractable decisions shows up in how the work is structured. A review of the Aircraft Recovery Problem finds that studies – and, by extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps together. They treat aircraft, crew, and passenger recovery as separable sub-problems, solved in sequence and handed off in turn [77]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage's constraints are applied. This is a documented instance of the myopic-optimization pattern this capstone investigates.

The pattern is visible even in work that improves on the status quo. The predictive turnaround-buffer allocation model of Kontodimou et al. [56] achieves a 37.1% reduction in propagated delay relative to a zero-buffer baseline, but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, outside the optimization. The gain is real within that scope. The framing also shows the risk this capstone is concerned with: a locally optimized decision layer can report a strong local benefit while passing tradeoffs to adjacent decision layers that its own cost model does not measure. The same pattern appears on the human side of the AOCC. Turhan, Gzara, and Elhedhli [89] document that when dispatcher workload exceeds capacity, dispatchers "may resort to conservative approaches, which can have financial implications for airlines" – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [36] frame this more broadly as a persistent gap between the disruption-management optimization literature and AOCC practice.

This report explores the value tradeoffs aviation stakeholders experience when a disruption occurs and the recovery decision is made with more or less myopia. A dispatcher or AOCC team member making a decision is often focused on its immediate effect – for example, minimizing delay for one flight – without visibility into what it means for the airline's wider operation or for the NAS as a whole. The cost models in use today capture the factors that are easiest to quantify, such as fuel and crew cost. Factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher and controller workload, environmental impact – are more often left out. The result can be congestion in particular airspace sectors, missed opportunities for more efficient routing, or lower profit driven by customer dissatisfaction. By modeling the NAS as a system of systems and tracing how operational intent propagates through it, the project aims to identify ways to improve these decisions and reduce the cost of disruptions to the air transportation system.

1.1 Overview of the Project

Plan ref: to-do-list.md S1 — Describe the industry/environment (commercial aviation and the NAS), the social context, and the key stakeholders (preview – full analysis in Section 1.3). State the project's research questions from to-do-list.md S1 ("Finalize project research questions") and explain why they matter. Draw on README.md's framing of the trajectory-intent lifecycle (Strategic objective → mission plan → flight plan → ATC constraints → trajectory negotiation → FMS intent → guidance commands → aircraft motion) as the throughline.

In the NAS, operational decisions are made by independent actors, each of which may or may not account for the effect on the others. Those actors include regulatory agencies, air traffic service providers, airport operators, airline operations centers, maintenance (MRO) suppliers, flight crews, passengers, and the aircraft systems themselves. Substantial architectural work exists within individual domains, but the relationships and information flows between domains are often incompletely

represented.

Existing models frequently focus on one actor or one problem, and show one or more of the following limitations:

- emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard decision-making;
- emphasis on aircraft systems without adequate representation of the operational context in which those systems operate;
- limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories;
- difficulty evaluating the effect of new operational concepts, automation technologies, and optimization services across organizational boundaries.

As a result, no broadly applicable architecture fully captures how strategic objectives become executable aircraft behavior. Without end-to-end traceability, engineers, operators, and decision makers have limited means to understand the consequences of a change within the NAS, and integrating future decision-support technology is harder than it needs to be.

This project addresses that gap by developing a System-of-Systems architecture that explicitly models the propagation of intent through the NAS, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

1.2 ~~Review of Current State Design~~ / Literature Review

Plan ref: to-do-list.md S2–S4 — Synthesize (do not merely list) findings from the three literature-review tracks: airline planning and operations (schedule planning, fleet assignment, crew scheduling, maintenance routing), day-of-operations and turnaround (A-CDM, weight and balance), and OCC/dispatch and flight execution (disruption management, dispatcher responsibilities, nominal ATC/IFR flow). Pull from per-source annotations captured via the literature-review workflow into `evidence/source-register.md` rather than re-deriving them here. Identify the research gap this capstone addresses (see also to-do-list.md S16 "Research gap").

Plan ref: to-do-list.md Track C (research gap) — What this project adds that prior work does not. The opportunity itself is documented: [15] measured what airline-supplied weight does for climb prediction; SESAR's ADS-C EPP downlinks the FMS's predicted profile; [19] gives the aircraft a picture of the traffic. State the three things this project does that those do not: (1) scores one decision at four stakeholder levels side by side (aircraft, airline, sector, system) with no weighted sum, so who pays is visible; (2) maps the decision to its authority and its information owner in a system-of-systems model, so the missing exchange is a named interface (`AirlinePerformanceShare`, D-019) and an unmet requirement (SLR-INF-02, D-018), not a finding in prose; (3) swaps the scoring model's fidelity under a fixed architecture (D-017), which is the MBSE claim.

Plan ref: to-do-list.md S11 (context for the experiment) — What the sector controller knows when resolving a conflict. Make the point that the controller has type, level, speed and route, but not weight or cost index. Cite [79] (p. 2, weight and speed intent "are considered competitive parameters by airlines and are not available for use") and [15] (one nominal weight per aircraft type). Say who does hold the weight (airline, FMS) and that they do not hold the authority.

Plan ref: to-do-list.md S11 (context) — What controllers do by default. Cite [73] (23 of 36 level-crossing cases resolved by a descent; controllers "use more predetermined rules"), [55] (leave the aircraft in steady state alone) and [34] (instruction counts per heuristic; "No point asking aircraft to climb if they come back and say they can't"). Note the disagreement between the first and third on whether descent is preferred.

This needs to be clear in my models.

Plan ref: to-do-list.md S11 (context) — TASAR as the mirror-image study. Cite [19]. Three points to make: (1) it gives the aircraft a picture of the traffic, where this project gives ATC a picture of the aircraft; (2) size of the effect per flight (27 gallons and 2.3 minutes for the typical case, range 12 to 103 gallons, p. 12); (3) it did not model aircraft weight and limited climbs to FL350 and below for that reason (p. 8). Point (3) is the assumption Section 3.2 adopts.

Plan ref: to-do-list.md S11 (context) — How sector loading reaches the controller. Explain the Monitor Alert Parameter in one or two sentences and what a "red sector" is. Cite [28] (§18-9) for the definition and [19] (p. 10) for the practice: the sector controller is generally not aware of red sectors elsewhere, and a supervisor may restrict traffic into one. Open: the report's "area manager" is not the order's term; red versus yellow is not defined in the order.

Plan ref: to-do-list.md S11 (context) – SOURCES NOT YET HELD — Aviation emissions and what they cost whom. Needed before Sections 4 and 5 can discuss emissions: (a) the CO2 produced per kg of jet fuel, with a citation (now held: [48]); (b) whether any carbon-pricing or offsetting scheme applies to a U.S. domestic flight, and a dated price; (c) a study of how emissions labels on flight listings affect passenger choice; (d) a social cost of carbon for the system-level discussion. Items (b) to (d) are not in `evidence/sources/`.

Time spent developing the Concepts of Operations also affected other work. For a period the project lacked a specific focus for which tradeoffs it would consider. The project is not an exploration of optimization, but its concept centers on sub-optimality driven by myopic decisions. The concepts of operations gave a more complete understanding of the NAS, the systems involved, and the interfaces among them, but they did not by themselves point to a specific scenario worth modelling.

The pivot from Phase 1 exploration to Phase 2 modelling was made to advance the project deliverables while building a deeper understanding of the information gaps. The intent was that when the modelling reached the measures of effectiveness and performance, the stakeholders' objectives would be revisited and a more targeted consultation of the literature could occur.

AI-DRAFTED, owner to rewrite — How the pivot helped (owner's note: "Talk about how this helped"). Drawn from `decisions/decisions-log.md` D-012, D-016, D-018 and D-019 and the journal. Check each claim against what you would say yourself.

That is what happened. Building the model forced questions the literature review alone had not: who owns the decision support for a trajectory change, which party holds the information that decision needs, and whether an exchange between them exists. Answering them located one specific gap. The sector controller decides which aircraft to move in a conflict, the airline and the flight management system hold the aircraft's weight and cost index, and no exchange carries one to the other. That gap is now a named interface in the model (`AirlinePerformanceShare`) and an unmet requirement, not a general statement about stovepipes. The literature consultation that followed was targeted in the way the pivot intended: sources on controller conflict-resolution practice, on climb prediction with airline-supplied weight, and on traffic-aware flight-deck requests were found and read because the model and the experiment needed them.

AI-DRAFTED, owner to rewrite — Current state of the project, October 2026. Replaces the struck mid-September paragraphs. Every number is from `task-board.md` and `journal/2026-10-11.md`; update them on the day of submission. Confirm what you want to say about state machines.

The project is now organized as four parallel tracks (model, experiment, report, and evidence on demand) worked in nine one-week turns that end on 2026-12-13. This plan replaced the single-pass August list on 2026-10-10. One traced thread carries the project: the airline's cost objective, the stakeholder and system requirements that follow from it, the airline-to-ATC interface, the conflict-resolution decision, and the experiment that scores that decision.

The requirements have been through two hand reviews and now stand at 13 stakeholder requirements and 23 system requirements, each limited to what bears on the studied decision. A third tier of 28 constituent-system requirements is drafted and not yet reviewed. The model holds the context

diagram, an aircraft internal block diagram that has been hand reviewed, an airspace management internal block diagram that is still a placeholder, a draft nominal-flight activity diagram, and the typed airline-to-ATC interface. The state machines the original plan scheduled for 9/15 have not been built.

The experiment is defined and runs end to end. Four aircraft in one en-route sector present one crossing conflict, and four cases resolve it with different information: a controller's rule, automation with one nominal weight per aircraft type, automation with the shared weight and cost index, and the same with connection information. An open aircraft performance model, OpenAP, scores every option. The first results are in Section 4.4 and are preliminary: ~~the scenario's numbers are not all confirmed.~~

1.3 Stakeholders

Plan ref: to-do-list.md S7 — Identify key project stakeholders (individuals/organizations affected by or perceiving themselves affected by NAS operational decisions). Draw on the PESTLE stakeholder discovery, stakeholder register, and RACCI matrix from to-do-list.md S7. Confirm the stakeholder set is diverse (airline, passenger, ATC/ANSP, airport, flight crew, environmental/societal, military – see S8's per-stakeholder objective categories) and that stakeholders with little formal power (e.g., passengers, communities affected by noise/emissions) are represented.

I created the `capstone_stakeholder_assessment.xlsx` file with an initial assessment of each candidate stakeholder's objectives, then refined it as I read more of the literature and understood the NAS better. For each stakeholder the assessment records:

- the stated interest, and whether it acts as a hard constraint, an optimization objective, a cost or penalty, a measure of performance (MOP), or a measure of effectiveness (MOE);
- candidate MOPs and MOEs;
- whether the stakeholder has a direct or indirect effect on trajectory decisions, if any;
- whether the stakeholder should stand alone or ~~be consolidated~~ abstracted through consolidation with another, and the reason.

Figure 1 shows the candidate stakeholders first identified, and Figure 2 places them by power and interest.

This led to the following list of stakeholders relevant to the project:

- Governments / Standards Organizations
- Operators (Airlines, Charter Operators, Cargo Operators)
- Air Traffic Control (ATC)
- Aircraft and Engine Manufacturers
- Airport Authority
- Flight Crew
- Passengers
- Environmental/Societal Organizations

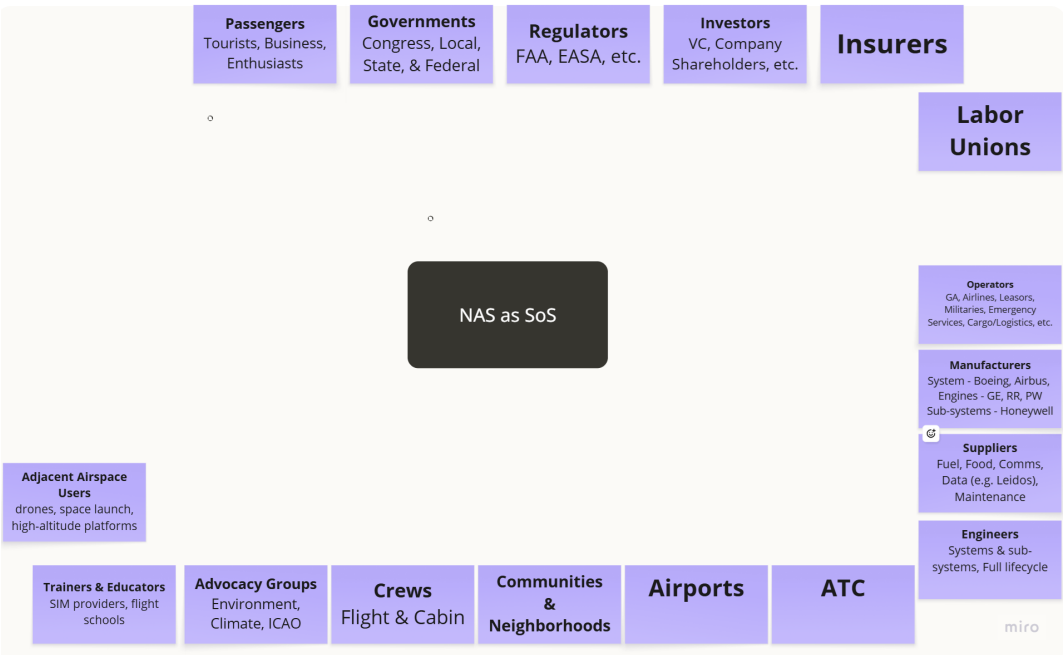


Figure 1: Candidate stakeholders of the NAS as a system of systems ~~initial exploration~~.

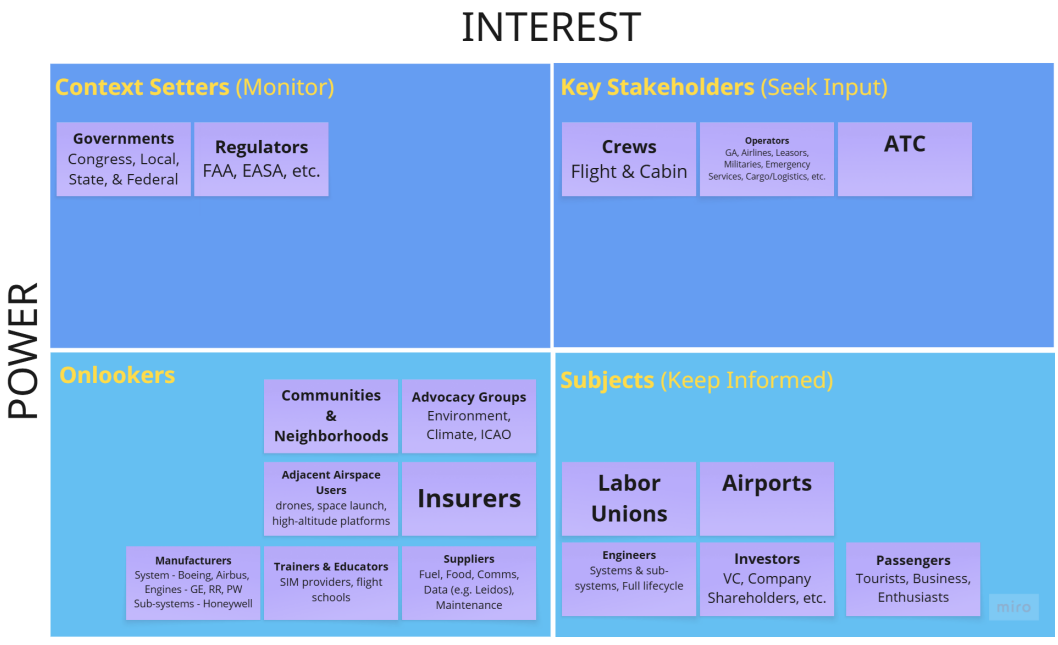


Figure 2: Candidate stakeholders placed by power and interest.

Talk about how context setters became part of the environment by defining the regulatory environment.

Describe how Airports were included b/c of streaming & capacity's relevance for decision making & representation in scenario was weak.



Figure 3: The NAS as a system of systems: candidate constituent systems (~~initial exploration~~).

Move to section discussing systems exploration

The aircraft is treated alongside these stakeholders as a constituent system. Figure 3 shows the systems first identified within the NAS, with the aircraft decomposed furthest, as its systems & their performance dictate the cost for each flight.

Move up near talk about xdsm.

An exploration of each stakeholder's interests led to the following list of objectives relevant to the project:

- Safety / Security
- Efficiency
- Throughput / Capacity
- Reliability / Resilience
- Environmental Impact
- Noise Impact
- Passenger Satisfaction
- Schedule Reliability
- Cost / Profitability
- Maintenance / Lifecycle Cost
- Regulatory Compliance
- Workload / Human Factors

1.4 Scope

AI-DRAFTED, owner to rewrite — De-scoping decisions and the final lists (owner's note: "Leave larger list above, discuss de-scoping decisions here and include the final list/diagram"). The reasons in quotation marks are your margin notes. Where your print struck an item without a reason, the reason given here is Claude's reading of the boundary criterion in Section 2.1 and is flagged in the list that follows this passage.

The lists in Section 1.3 are the result of exploration. The study itself is narrower. It examines one tactical decision, the resolution of a conflict between aircraft in cruise, and keeps only the stakeholders and objectives that bear on that decision.

Map to Context diagram names - cite figure.

Four stakeholders remain, together with the aircraft: operators (the airline), air traffic control, the airport authority, and the flight crew. Governments and standards organizations, aircraft and engine manufacturers, passengers, and environmental and societal organizations are outside the System of Interest. They supply rules, equipment, demand, and expectations to the decision but do not make it or execute it, so they are treated as boundary actors and not decomposed.

stakeholder

Six objectives remain: safety, efficiency, environmental impact, schedule reliability, cost, and workload. Schedule reliability is carried in the experiment as a step in cost when a flight's delay passes a threshold (the parameter K of Section 3.3). The others were set aside for the following reasons. Noise impact: the measure is "not relevant at cruise." Passenger satisfaction: "not measurable objectively." Security, ~~profitability, reliability~~ and resilience, maintenance and ~~lifecycle cost,~~ and regulatory compliance are not changed by the choice between resolution options in the scenario.

Figure 4 is the resulting System of Interest as modeled: the constituent systems kept and the exchanges among them.

Profit is fall out of other objectives, tracked at operator organizational level & is more complex than any one decision. Reliability captured w/ schedule reliability. Maintenance & Lifecycle cost are largely determined by # of flight cycles, chosen takeoff performance, flight distance, & climb controls. ATC decision will have negligible impact.

Plan ref: to-do-list.md S1 (owner's open question, 2026-10-11 print, page 6) — "Why am I ignoring throughput/capacity, which would benefit from an improved ACM?" Answer this here before throughput/capacity is listed as out of scope: either give the reason it is set aside for this study, or bring it back as an objective and say how the experiment would measure it. Also confirm or replace the reasons Claude supplied above for the four stakeholders and for security, profitability, reliability/resilience, maintenance/lifecycle cost and regulatory compliance: your print struck these without a note.

Throughput & Capacity were left for another study. Might also be worth mentioning that they're close to ongoing work by the author, so an abundance of caution lead to avoiding that aspect of problem space.

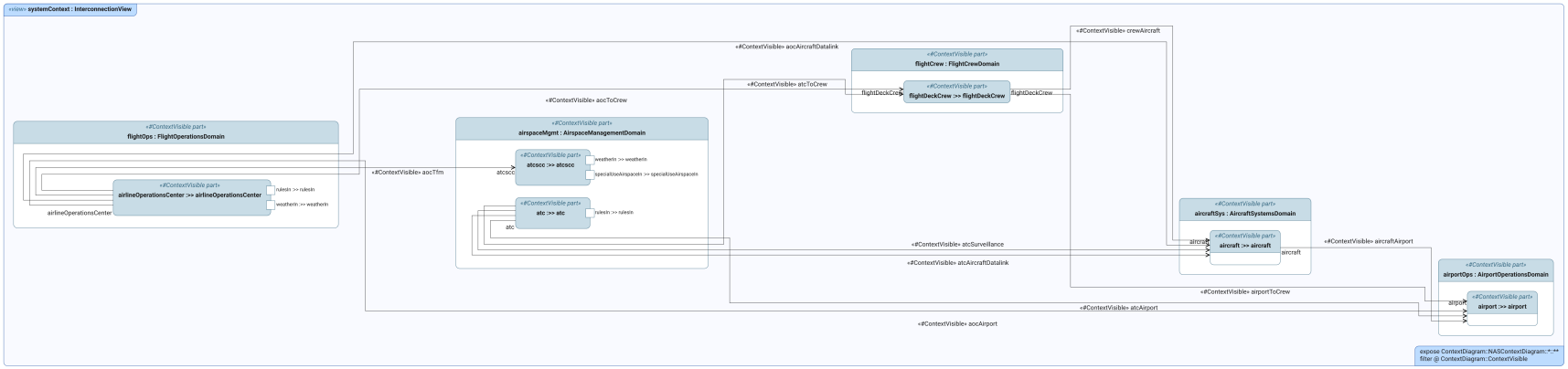


Figure 4: NAS context diagram. View `systemContext`; source `nas_context_diagram.sysml`; rendered 2026-10-10.

1.5 Project Deliverables

Plan ref: to-do-list.md S10, S14, S16 — State the goals and deliverables: the SysML reference architecture, the traceability chain from stakeholder needs to aircraft-level behavior, and the experiment (Section 3). Include the quantifiable measures used to assess success (e.g., traceability coverage, stakeholder coverage, demonstration of a local-vs-SoS tradeoff).

The following deliverables were established for the project at its inception: Section 1.2 describes the current status of these deliverables.

Context, body, ibd, activities, requirements, experiment & traceability chain & gap analysis

1.6 Project Timing

Plan ref: to-do-list.md S1 (new) — Revisit the timeline from the 503 Proposal submittal (see `prework/ISD 503 Submittal.pdf`) and show the plan now in force: the nine one-week turns of D-016 (the table at the top of `to-do-list.md`), each with its exit test and target date, next to the August milestones in Table 1.

Have Claude propose a table

2 Assumptions and Methodology

expand upon.

The problem is that the inertia of past, piecemeal designs and the high cost of upgrading the NAS make the system hard to change. The project proposes a new airline-to-ATC interface and a new way of deciding which aircraft to move in a conflict, which together could lower the cost of air traffic management.

Toward those ends, the project uses a model-based systems engineering (MBSE) approach to design the architecture and the decision-making process, and a simulation-based experiment to test the proposed interface and decision-making process. First, a literature review was conducted to:

1. identify the current state of air traffic management,
2. identify the relevant stakeholders and their objectives,
3. identify the requirements necessary to meet those objectives, and
4. identify the gaps in the current system that the proposed interface and decision-making process can address.

2.1 Systems Engineering Approach

which produce

Standard systems engineering methods of analysis and decomposition were applied to the collected evidence, producing a context diagram, internal block diagrams, and activity and sequence views. These identified the relevant systems and the interactions among them that are necessary to meet the stakeholders' objectives. From them the stakeholder requirements were written and then decomposed into system-level requirements (SLRs) for the NAS. The SLRs were allocated to the relevant constituent systems, and high-level requirements (HLRs) decomposed them one level further.

This enabled the project to (1) build a model of the architecture and the decision-making process, and (2) compare current air traffic management practice, which relies on general rules of thumb and knowledge of what has worked before [73, 55], with a model-based decision-support approach that replaces the low-fidelity decision rule with one or more higher-fidelity models of aircraft performance, at varying levels of accuracy. & information.

The emphasis is on structure, behavior, interfaces, and traceability, not on optimization mathematics. The project's commercial value to the author lies in a better understanding of the systems,

OLD plan Presented for reference. Disposition each's status w/ not started, draft, complete, or will not implement.

Table 1: Project milestones and verification methods

Project Milestone	**Deliverable**	**Lead/Team Member**	**Expected Completion Date**	**Target Metric/Verification Method**	
Literature Review	Annotated bibliography	Owen	9/7/2026	10–20 papers categorized	draft
PESTLE Analysis & Systems-of-Interest Definition	Context diagram, text-based documentation	Owen	9/10/2026	Review	Complete
Systems-of-Interest Definition	Text-based documentation	Owen	9/15/2026	Clear boundaries and exclusions	Complete
Operational Scenarios	ConOps document	Owen	9/22/2026	1–3 representative scenarios	draft
Draft Architecture	BDDs and package structure	Owen	10/1/2026	All domains represented	Complete
Interface Definition	Interface Control Document	Owen	10/12/2026	All interfaces identified	draft
Internal Block Diagrams	IBDs	Owen	10/17/2026	Critical interfaces connected	draft
Requirements Definition	Requirements model	Owen	10/20/2026	Trace to stakeholders	draft
Measures of Effectiveness	MOE/MOP hierarchy	Owen	10/25/2026	Quantifiable metrics defined	Complete
Requirement Traceability	Requirements matrix	Owen	10/29/2026	Complete allocation	draft
Activity Diagrams	Operational activities	Owen	11/3/2026	Scenario walkthrough	draft
Sequence Diagrams	Information exchanges	Owen	11/8/2026	Message completeness	N.S.
State Machines	State behavior of key entities	Owen	11/13/2026	Executable state logic	WNI
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed)	Owen	11/17/2026	All decisions assigned	WNI
Optimization Prototype	Working code	Owen	11/21/2026	Unit test cases	draft
Integration into MBSE Architecture	Traceability links	Owen	11/24/2026	End-to-end scenario	
Final Baseline & Final Paper	Versioned model & thesis/paper	Owen	12/16/2026	Draft	
Paper Submittal	Thesis/paper	Owen	12/20/2026	Official submittal	

their interactions, and the divisions of responsibility that inhibit information flow and lead to sub-optimal outcomes. To explore that in a meaningful way, decision support and optimization are treated as a capability inside the architecture, which following MBSE principles makes possible.

More detailed documentation of the systems and their constituent systems is possible (examples are in Appendix A.3), and more formal mathematical models could be used. That effort was judged to be outside the work required to deliver the commercial value to the customer. In keeping with customer-focused MBSE, the project's artifacts are tailored to the customer's needs.

Individual steps in the process were likewise tailored to the author's availability. After an initial literature review, the review continued ad hoc as the work required. A full gate-to-gate concept of operations was constructed informally and refined with publicly available general knowledge of the NAS. No alternative system decompositions were formally explored beyond the initial abstraction decisions. In this way the ISO/IEC/IEEE 15288 technical processes were followed, including the tailoring of those processes to the application that the standard provides for.

Table 2 summarizes the systems engineering processes and the artifacts that realize them.

Plan ref: to-do-list.md Track C (owner's note, 2026-10-11 print, page 10) — "Check these process steps in the actual doc. Itemized?" The process names in the first column of Table 2 were written from memory of ISO/IEC/IEEE 15288 and the INCOSE handbook. The standard is not held (`references.bib` has a web-only entry). Check each name against the standard's own list of technical processes, and decide whether the table should itemize all of them and mark the ones tailored out.

Table 2: Systems engineering processes and the artifacts that realize them.

Process (15288 / INCOSE)	Artifact in this project	Where
Stakeholder needs and requirements	13 stakeholder requirements, each traced to an objective row	`requirements_stakeholders.sysml`; Appendix
System requirements	23 system requirements, baselined or improved (D-018), allocated to constituent systems; 28 constituent-system requirements one tier down (drafted, not yet reviewed)	`requirements_nas_system.sysml`; `requirements_constituent_systems.sysml`
Architecture definition	Context diagram, internal block diagrams, typed interfaces on the thread (D-008), activity and sequence views	`cameo_models/`; Section 4.3
System analysis	Experiment at three fidelities (D-007), outcome table	`simulation/`; Sections 3 and 4.4
Verification	Verification cases for the improved requirements, run by the experiment	`nas_verification.sysml`
Validation	Stakeholder-level reading of the outcome table (who pays)	Section 4
Technical measurement	MoEs (outcome-table columns), MoPs (requirement measures)	Section 3.3
Configuration management	Decisions log, model text as the authoritative source (D-006), model freeze	`decisions-log.md`
Tailoring	D-016	this subsection

Plan ref: to-do-list.md S10 — Describe the MBSE methodology used to build the architecture in SysML v2 (textual notation authored in Syside Modeler in VS Code, with diagrams as generated views): package organization, view types produced (BDD, IBD, activity, sequence, requirements), and the traceability approach (stakeholder needs → objectives → requirements → systems → activities → information exchanges → aircraft behavior). Note per `workflows/update-architecture.md` (decision D-006) that the SysML v2 text is the source of truth; Cameo is not used as the source of truth. Describe the working loop (draft, print, hand review, transcribe, apply, log, re-render) and the artifact states 0 to 4 (D-016).

Plan ref: to-do-list.md S6 — Decomposition kept, alternatives named. One paragraph: why the authority / responsibility / information-ownership decomposition (D-002, D-007) was kept, naming the alternatives (organization-based, lifecycle-based, physical, information-flow, decision-authority) and the criterion (which one makes the decider, the information owner and the authority visible on the same picture). Building the alternatives is future work (D-016).

WNI

2.2 Working Assumptions

The working assumptions fall into two groups: assumptions about the problem and its context, and assumptions about the methodology.

The problem statement assumes that:

- the abstractions used in the model are sufficient to capture the relevant behavior of the system;
- the stakeholders' objectives are comprehensively captured and accurately represented in the model.
- The scenario is sufficient to explore the potential benefit of interconnected NBSE

Every abstraction was made with the express goal of capturing the relevant behavior of the system, as defined by the stakeholders' objectives. The System of Interest, the NAS as a system of systems, was ratified around the trajectory-intent lifecycle, from enterprise objectives through aircraft motion. An entity is modeled when it has distinct decision authority or execution behavior that is dynamically coupled to in-scope decisions and materially affects cost, safety, workload, schedule reliability, or passenger value. Boundary actors supply objectives, demand, feedback, or rules without being decomposed as decision mechanisms. Exogenous influences enter as constraints or scenario parameters. This criterion preserves the authority and responsibility relationships a system-of-systems architecture needs without expanding the project into a model of every NAS participant.

The methodology ~~does not assume that the proposed methods are better; that is what the experiment tests. It~~ assumes that:

- models of every required aircraft type are available;
- the security concerns of sharing airline data can be mitigated;
- the proposed interface can be implemented in the NAS;
- digital communication channels exist between the airline and ATC; ∄/or the A/c & ATC
- the specific scenario presented in the experiment is representative of a larger set of scenarios that generalize to the NAS as a whole.

Combine. Later cite EFP; value for CL&W, BADA4 for API models?

AI-DRAFTED, owner to rewrite — ~~Limits of the experiment, moved here from the Results section (owner's note: "Add to assumptions"). The items are those of the old "Limits" call-out. The misreporting figure has no registered source yet.~~

~~The last of these assumptions carries the most weight, and the following limits bound it:~~

Add to discussion instead of the list.

- Conflicts between two aircraft crossing at the same level are a minority of conflicts: 36 of 256 in the operational data of Rantanen and Wickens [73].
- The experiment uses one invented en-route sector and four aircraft.
- The conflict is a static snapshot; the traffic does not evolve while the decision is made.

reword & add to primary list.

- The rule that limits climbs when the weight is unknown is a single threshold (FL350) applied to every aircraft type. Cite source.
- The airline is assumed to report its weight and cost index truthfully. An airline could misreport what it shares.

Accurate info. Move to primary list.

The simplifications inside the aircraft performance model are listed separately in Section 3.4.

Plan ref: to-do-list.md Track C — Misreporting. Cite the 1 to 3 percent figure for the last limit once that source is registered in `evidence/source-register.md`.

Move to discussion Regarding accuracy bullet, above.

2.3 Proposed Airline-to-ATC Interface

This should it be its own section here. Where would this fit?

Plan ref: to-do-list.md Track C (D-019) — Name the digital-thread artifact. Describe `AirlinePerformanceShare`: what it carries (gross weight, cost index, connection priority; thrust, drag and fuel-flow model identifiers present but not exercised), who sends it (airline operations center or FMS), who receives it (ATC decision support, per D-012 owned by the system that decides), and which requirements it satisfies (SLR-INF-02) and the goal it serves (SLR-CLR-13). State that security is an assumption. **Owner's note, 2026-10-11 print: "important to do"** — position it against ADS-C EPP and Coppenbarger's Table 2 (the same items, a different receiver); a first draft of that positioning follows.

AI-DRAFTED, owner to rewrite — Positioning against prior work (the item you marked "important to do"). The Coppenbarger and Schultz statements are from `decisions/decisions-log.md` D-019, which cites Table 2 and page 2 of those papers. The ADS-C EPP sentence rests on `scsarju2019solutionsCatalogue`, registered from its title page and contents only: read the entry for the solution before keeping the sentence.

The items the interface carries are not new. Coppenbarger [15] listed take-off weight, thrust and drag factors, and the speed profile as candidates for exchange between an airline and air traffic control, and measured what airline-supplied weight does for the prediction of a climb. In Europe, the Extended Projected Profile is sent over ADS-C to downlink the trajectory that the flight management system predicts [82]. What has kept the items out of use is ownership, not feasibility: weight and speed intent "are considered competitive parameters by airlines and are not available for use" [79].

Also availability of 1st principles data (T, D, FF)

The proposal here keeps the items and changes the receiver and the purpose. In the earlier work the data feeds a trajectory predictor, to make the prediction of one aircraft's path more accurate. Here it feeds the decision support of the party that chooses which aircraft to move, to change the choice. The interface is typed in the model, tagged as part of the improved architecture, and tied to the requirement it satisfies, so the report can point at a model element where the earlier work has a finding in prose. Confidentiality is stated as an assumption: the data is disclosed only to the deciding party, and no security mechanism is modeled.

This provides for what-if-ing w/in a mathematical framework.

3 Experiment

Plan ref: to-do-list.md S11 — State the central claim the experiment explores. Still undecided (see `experiment-definition.md` §1 and the hand review notes 1.2, 2.1, 4.1). Answer the objection that the simulation compares two ways of operating and so does not by itself test MBSE.

3.1 Scenario

Plan ref: to-do-list.md S11 — Scenario. Four aircraft in one invented en-route sector, shown as one snapshot: a crossing pair at the same level and two aircraft in trail 2,000 ft below. State which aircraft belong to the same airline, which is late with connections, and the weight of each. Give the level of the crossing pair and say why it is above FL350. Justify the static snapshot by the controller studies that used one ([55], [34]).

Plan ref: to-do-list.md S11 (owner's note, 2026-10-11 print, page 11) — "Justify one scenario (time) and expand as possible." Say that one scenario is studied because of the time available in one term, name what a second scenario would add, and list the variations that are run on the one scenario (similar weights, the climb closed, a crossing pair at or below FL350).

Plan ref: to-do-list.md S11 (found while placing the figure) — The drawing and the scenario file disagree on one number. Figure 5 is labelled "20 min to WPT"; `simulation/scenario_base.yaml` has `time_to_crossing_min: 12`, meaning the decision is made 12 minutes before the two aircraft reach the crossing. The scoring code does not read that value, so no result changes whichever is kept. Decide which is right and change the other.

This is ok for now.

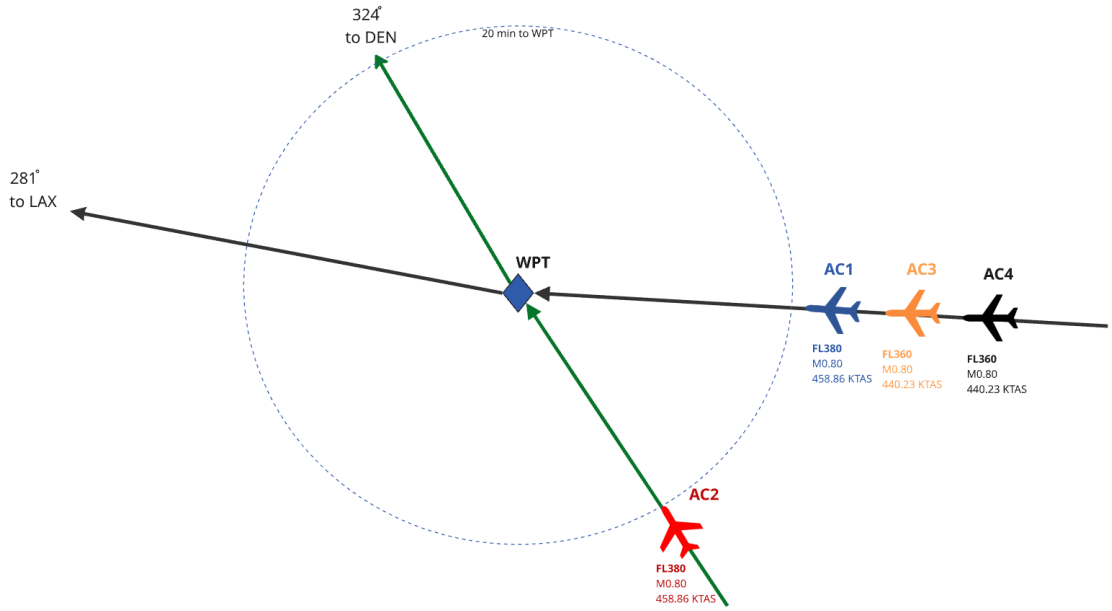


Figure 5: The scenario in plan view. AC1 and AC2 cross at the waypoint at FL380; AC3 and AC4 follow AC1's track at FL360.

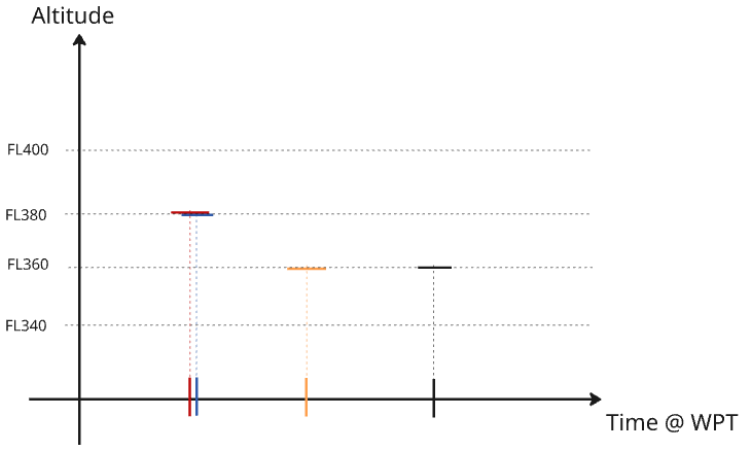


Figure 6: Level of each aircraft against its time at the waypoint. AC1 and AC2 arrive together at FL380.

3.2 Cases and Resolution Options

Plan ref: to-do-list.md S11 — The one thing that differs between cases. Define cases A to D by what the decider knows. State what "automated decision" means for who devises and who issues the clearance (undecided). State that the levers are the same in every case.

Table 3: The four cases. Wording to be the author's.

Case	Decider	Information	Climb rule applies?
A	Controller, by rule	Type, level, speed, route	Yes
B	Automation	Nominal weight per type	Yes
C	Automation	Shared weight and cost index	No
D	Automation	Shared weight, cost index and delay margin	No

Plan ref: to-do-list.md S11 — Altitude options and the climb rule (assumption adopted from a prior study). State the three altitude options (2,000 ft above, 2,000 ft below, 4,000 ft below) and the rule that a climb is offered only at or below FL350, citing [19] (p. 8). Say plainly that this is that study's modeling assumption, made because it did not model aircraft weight; it is not an FAA rule and not observed controller behaviour. Then explain why the rule binds cases A and B and not cases C and D: this is the mechanism by which missing information removes an option.

Plan ref: to-do-list.md S11 — Baseline rule for case A. One descent clearance, citing [73]. State which aircraft descends and the tie-breaker (undecided), and whether the aircraft is cleared back up.

Plan ref: to-do-list.md S11 — Resolution options. Introduce Table 4. For each option give the instructions it costs now, the follow-up it creates, and what the decider must know to use it. Cite [34] for the instruction accounting.

Table 4: Resolution options. A dash means no instruction. Full version with instruction counts in Appendix A.5.

Option	AC1	AC2	AC3	AC4
O1	Descend 2,000 ft	–	–	–
O2	–	Descend 2,000 ft	–	–
O3	Climb 2,000 ft	–	–	–
O4	–	Climb 2,000 ft	–	–
O5	Turn behind AC2	–	–	–
O6	–	Turn behind AC1	–	–
O7	Reduce speed	–	–	–
O8	Descend 4,000 ft	–	–	–
O9	Descend 2,000 ft	–	Slow or turn	–

Plan ref: to-do-list.md S11 — Sources for the option set and the counts (owner's note, second hand review, 2.4: "Add these to report!"). Levers: `experiment-definition.md` §4, and [19] (p. 8) for lateral and altitude changes; that study does not use speed. Altitude steps and climb rule: [19] (p. 8), with the authors' reason. Instruction counts: [34] (pp. 72–74). Climbs used less than descents: [73] (6 of 36). Speed limited to ±0.04 Mach and built to fail: the author's own figure, to be stated as experience.

Plan ref: to-do-list.md S11 — The baseline is not hobbled (owner's note, second hand review, 3.2). In the later variation at or below FL350 the baseline may climb and does not, because its rule is one descent. Say that this matches the observed preference for descents ([73]). **Owner's note, 2026-10-11 print: "Also note higher alt ∼ optimal"** — say that the higher level is near the optimum for these aircraft (Figure 13), so a baseline that never climbs is giving something up.

3.3 Scoring

Plan ref: to-do-list.md S11 — What the system column is (owner's notes, second hand review, 5.5 and 6.1). State that it is a measure of the airspace as a whole system of systems, define it by its cost statement (total fuel, total CO2, total airline cost, workload items, largest cost to any one aircraft), and refer back to that statement wherever the column is discussed.

Plan ref: to-do-list.md S11 — How each option is scored, one statement per viewpoint. Aircraft: extra fuel plus cost index times extra time ([69], Eq. 1–3). Airline: the sum over its aircraft plus a step cost when a flight misses a connection, plus CO2 converted to dollars. Sector: clearances issued, effort to devise them, and residual risk, reported as three separate items. System: totals stated side by side, with no weighted sum (D-013). Say why no weighted sum: nobody in the model owns the weights.

Plan ref: to-do-list.md S11 — Emissions. State the conversion from fuel to CO2 and its source ([48]). State how CO2 becomes dollars at the airline level and why that route was chosen (credit price, passenger value, or both; undecided), and why the system level keeps CO2 in kg.

For aircraft i the cost of an option is its extra fuel plus its cost index times its extra time, in kilograms of fuel, the form used in [69]:

$$c_i = \Delta m_{f,i} + CI_i \, \Delta t_i, \tag{1}$$

where the extra fuel $\Delta m_{f,i}$ and the extra time Δt_i are the option minus the baseline of flying on undisturbed, and CI_i is the cost index of aircraft i in kilograms of fuel per minute.

Each automated case chooses, among the options it is allowed to offer, the one with the lowest estimate

$$J = \sum_i \left(\Delta m_{f,i}(\hat{m}_i) + \widehat{CI}_i \, \Delta t_i(\hat{m}_i) \right), \tag{2}$$

where the hats mark what the case believes. Case A does not use Eq. 2; it applies its rule, one descent. Case B uses the type's nominal mass for every aircraft and values time at 1 kg/min. Cases C and D use each aircraft's actual mass and cost index. Case D also rejects an option that delays a flight with connections past its threshold. Whatever a case chooses is then scored at the actual masses, and that score is what the tables report.

Schedule reliability enters as a step: a flight with connections that is delayed past its threshold incurs a cost K on top of Eq. 1. No sourced value of K is held, so the tables report whether the threshold is crossed, not a dollar figure. Carbon dioxide is 3.16 kg per kilogram of fuel [48]. Instructions are counted, now and later, and are not converted to anything.

Plan ref: to-do-list.md S11 (candidate figure) — Figure 7 is another paper's figure. It is Figure 1 of [72], placed here because it shows the step in the cost of delay that K stands for. Decide: keep it as a reproduction with the credit in the caption, redraw it for this scenario, or remove it.

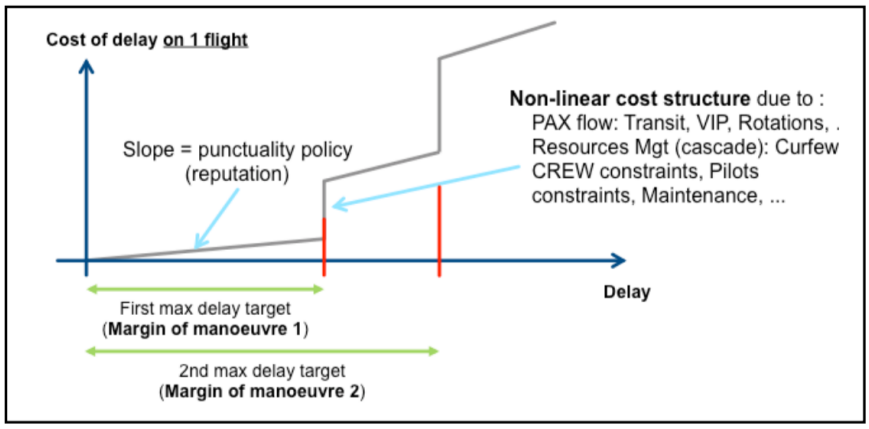


Figure 1: Typical cost-delay model profile per flight

Figure 7: Cost of delay for one flight, with steps where a margin is exceeded. Reproduced from [72], Figure 1.

3.4 Aircraft Model and Its Simplifications

Plan ref: to-do-list.md S11, S12 — Truth model (decided, D-017). OpenAP (open aircraft performance model, TU Delft) scores every option: fuel flow, thrust and drag for the B737-9 at the aircraft's actual mass. State the three selectable fidelities (conceptual: feasibility and instruction counts only; nominal mass; actual mass) and that the decider's case (A to D) and the truth model's fidelity are independent settings. The earlier kinematic model (`vehicle_dynamics.py`) is not used by the experiment.

Each aircraft is a point mass in steady flight at a constant Mach number. Drag, maximum thrust and fuel flow are taken from OpenAP [88], which uses a two-term drag polar (its Eq. 30 and 31) and, above 30,000 ft, the two-shaft turbofan thrust model of Bartel and Young [3] reproduced as its Eq. 22 and 23); those equations are in the OpenAP paper and are not repeated here. The fuel-flow curve is the exception: the installed version of the package (2.6.2) uses a different curve from the one the paper describes, so for fuel flow the paper is the source of the data (the ICAO engine emissions databank) and the package is the source of the curve. Table 5 lists the values OpenAP holds for the type. Aircraft mass enters through drag: a heavier aircraft needs more lift, and its drag rises with the square of the lift coefficient.

Plan ref: to-do-list.md S11, S12 (owner's note, 2026-10-11 print, page 13) — "Cite another paper for this content. Don't re-create." The OpenAP paper is now held (`evidence/sources/sun2020openap_aerospace-07-00104.pdf`) and was read in full by Claude on 2026-10-11; the equation numbers above are from that reading. The author has not read it. Bartel and Young is not held, and is cited here only through the OpenAP paper, which reproduces its equations. Read Sections 5 and 7 of the OpenAP paper (pp. 6 to 17) before submission. What the paper says about its own accuracy is in `evidence/literature-notes/summaries/4 - sun2020openap.md` and belongs in the discussion of the truth model's uncertainty (Section 4.4).

A descent is flown at 1,500 ft/min. A climb is flown at maximum thrust, at the rate that thrust allows and no more than 1,500 ft/min (2,024). A level is available to an aircraft only if it would still have 150 ft/min of climb left there. Cases A and B do not know the mass and so cannot make that test; they offer a climb only at or below FL350. Cases C and D make it at the aircraft's actual mass.

Figure 8 is drawn assuming lift equals weight. Its low-speed boundary is 1.3 times the stall speed at an assumed clean maximum lift coefficient, and its high-speed boundary is the maximum operating Mach number. The aircraft's present level and Mach number are marked, and each

Table 5: Values OpenAP 2.6.2 uses for the B737-9 (type code B39M), read from the installed package on 2026-10-11.

Quantity	Value	Note
Wing area S	124.6 m^2	type data
Zero-lift drag coefficient C_{D0}	0.020	737 MAX 8 polar, borrowed
Induced drag factor k	0.042	737 MAX 8 polar, borrowed
Engines n, type	2, LEAP-1B	type data
Static thrust per engine T_0	111,300 N	engine data
Reference thrust T_{cr}	46,300 N (both engines)	$0.2\,T_0 + 890$ N per engine
Reference cruise M_{cr}, altitude	0.79, 11,000 m	type data
Take-off fuel flow s	0.877 kg/s per engine	ICAO databank
Fuel curve c_1, c_2, c_3	0.9376, 1.9768, 1.3955	package default curve, not fitted to this type

altitude and speed instruction the scenario offers it is marked as inside or outside the region with 150 ft/min of climb left.

The model makes the following simplifications.

1. **Standard atmosphere, no wind, lift equal to weight.** Time differences come only from true airspeed and added track distance.

2. **No compressibility drag, ~~and the climb left is sensitive to it~~.** The drag used here has no Mach-number term. The OpenAP paper does include one, a wave-drag term above a critical Mach number, and says the effect "should not be ignored" there [88] (Section 5.1); the installed package leaves the term off unless asked and marks it experimental, and the experiment runs with it off (D-024). The FAA classes flight from Mach 0.75 to 1.20 as transonic [22] (Para. 1-4e), so the scenario's Mach 0.80 is at the low end of that range. With OpenAP's experimental wave-drag term switched on, drag at Mach 0.80 is 2 to 4 percent higher. That is 1.3 kN for the heavy aircraft at FL380, against 2.7 kN of excess thrust, so its climb left falls from 181 to 93 ft/min, under the 100 ft/min of the service ceiling. The light aircraft at FL400 falls from 322 to 239 ft/min and still clears 150. The weights in the scenario were set with the term off. Not decided: whether to switch the term on and set the weights again.

3. **The drag polar is the 737 MAX 8's.** OpenAP has none for the 737 MAX 9 (D-017).

4. **The fuel curve is the package default.** OpenAP has no fuel model fitted to the 737 MAX 9 or MAX 8; it uses its default coefficients scaled by the LEAP-1B take-off fuel flow.

5. **The climb margins.** The service ceiling is the level with 100 ft/min of climb left [22]. The 150 ft/min the experiment requires is the margin the author has seen airlines keep for clear air turbulence and other unknowns (industry experience, no document held; D-023).

6. **One climb rate per climb.** The rate is taken at the mid-altitude and held; the climb is not integrated. A climb is not checked against the time left before the crossing; the longest in the base scenario is the heavy aircraft's 4,000 ft climb back after option O8, about 12 min, which happens after the conflict.

7. **Mass is constant.** Fuel burned during the manoeuvre does not reduce the mass.

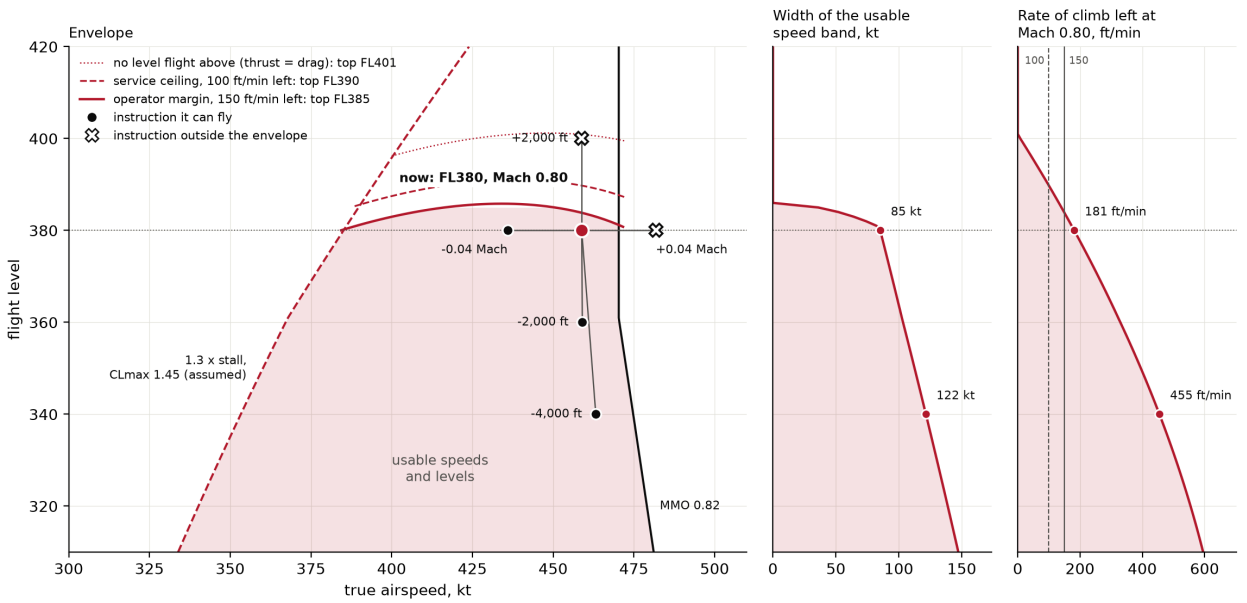


Figure 8: The heavy B737-9 (71.0 t) has little room left to change speed or level at FL380. Left: cruise envelope, with three lines of constant rate of climb left: 0 ft/min (thrust equals drag), 100 ft/min (the service ceiling, [22]) and 150 ft/min (the operator margin the scenario uses). Right: width of the usable speed band and rate of climb left at Mach 0.80, by flight level.

8. **The low-speed limit rests on an assumed maximum lift coefficient** ($C_{L\,\max} = 1.45$) and is a stall margin, not a buffet boundary.

9. **A speed change on the crossing pair is not calculated.** It is marked unable to separate the pair by construction (D-014).

Plan ref: to-do-list.md S11 — Abstractions made and why. One short table: each simplification, the reason, and the risk. A draft list is in `experiment-options-and-scoring.md` §6.

3.5 Worked Example: Option O4, the Light Aircraft Climbs 2,000 ft

The light aircraft, AC2, is at 62,980 kg, FL380 and Mach 0.80. At FL400 it would have 322 ft/min of climb left, more than the 150 ft/min required, so the level is available to a case that knows the mass.

At maximum thrust it climbs at 409 ft/min, so the 2,000 ft take 294 seconds. OpenAP gives a fuel flow of 0.5572 kg/s at FL380, 0.6321 kg/s in the climb, and 0.5465 kg/s at FL400. The aircraft then holds FL400 for 15 minutes (900 seconds). The extra fuel is what the climb costs less what the higher level saves:

$$\Delta m_f = (0.6321 - 0.5572)(294) + (0.5465 - 0.5572)(900) = 22.0 - 9.6 = 12.4 \text{ kg}.$$

The extra time is zero: at these levels the standard temperature is constant, so Mach 0.80 is the same true airspeed at FL380 and FL400. With no extra time, the cost index does not enter, and the cost of the option to AC2 is 12.4 kg. This is the 12 kg, rounded, in the case C row of Table 6.

4 Results and Discussion (Preliminary)

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Results and Discussion (preliminary). For Interim Report #2 this section needs: (1) the architecture as it stands (context diagram, aircraft IBD, the three requirements tagged `improved` as the measured gap) — the figures are now in Sections 1.4 and 4.3; (2) the two hand checks from `experiment-definition.md` §9 written up as preliminary results; (3) the first OpenAP numbers — Figure 13 and Table 6 are in place, the discussion is not; (4) one paragraph on what is not yet done.

4.1 Operational Concept

AI-DRAFTED, owner to rewrite — The operational concept: full detail, then the abstraction and its reason, then the example conflict (your notes: "DO"; "Show full detail 1st, then w/ abstractions and provide rationale"; "Need to update w/ the example conflict"). Sources: `knowledge/models/conops-nominal-domestic_flight.md`, `knowledge/models/conops-hub-to-hub-trajectory-cost.md` (both marked as working documents, not reconciled against the literature), D-016 item 5, and the activity diagram. `FlyEnroute` is not yet decomposed in the model, so the last paragraph describes where the conflict will sit, not what the diagram shows today.

The operational concept was first written in full detail as a gate-to-gate thread for one nominal commercial flight, from the pilot's point of view. It runs through eleven phases: the airline's schedule, fleet and crew assignment; the pilot's preparation before duty; reporting and the dispatch release; aircraft preflight and initialization of the flight management system; boarding, loading and fueling; clearance, pushback and taxi; takeoff and initial climb; climb; cruise; arrival planning; and descent through to the gate. Each phase records its initial state, the interfaces in use, and the questions it raised. The full thread is in Appendix A.1.

The model carries an abstraction of that thread. Figure 9 shows six actions in sequence: plan the flight, prepare for departure, depart, fly en route, arrive, and close out the flight. The eleven phases were reduced to six because the study concerns one decision in cruise. The phases before departure and after arrival fix the conditions that decision inherits (the aircraft's weight, its cost index, its schedule position) and are kept at the level needed to show where those conditions come from. Detail is added only where the studied decision is made.

The example conflict sits inside the fly en route action. Two aircraft in cruise are predicted to cross at the same level at the same time. The sector controller must move one of them, and each choice costs fuel, time and instructions that fall on different parties. The nominal thread treats a trajectory change in cruise as one step; the experiment of Section 3 opens that step and asks what the decider knows when making it.

Plan ref: to-do-list.md S5 — Decompose `FlyEnroute` one level in `nas_activity_diagram-nominal_ops.sysml` so the conflict-resolution step appears on Figure 9 (closes item 5), then check the three paragraphs above against the diagram. Put the full activity breakdown in Appendix A.1.

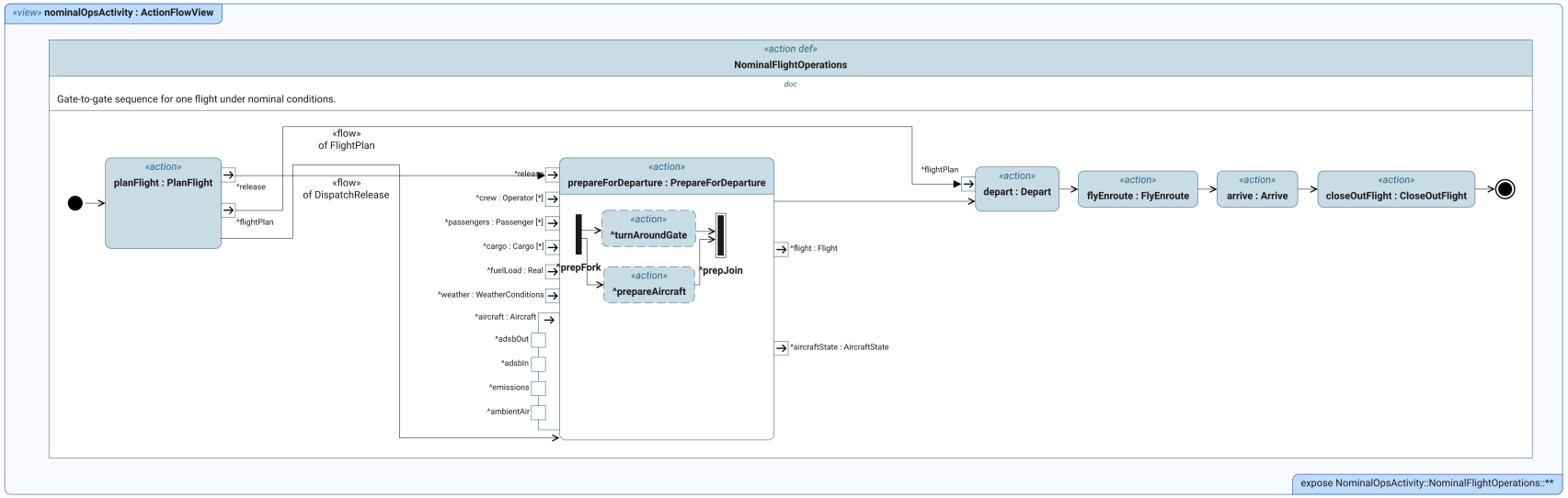


Figure 9: Nominal flight operations activity diagram (draft: no swimlanes, sub-actions are placeholders). View `nominalOpsActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-10.

Move to earlier section that discusses objectives?

4.2 Stakeholder Objectives

AI-DRAFTED, owner to rewrite — What "optimal" means to each stakeholder, and where the objectives conflict (your note: "DO"). Source: the synthesis section of `knowledge/models/stakeholder-objective-ontology.md`, whose own status note says the classifications are judgment calls not yet sourced from annotated literature. Limited here to the stakeholders and objectives kept in Section 1.4.

Each stakeholder kept in scope pursues a different objective when an aircraft's trajectory is changed. The airline wants the lowest cost for its flights, which combines fuel, time, and the consequences of a late arrival for connecting passengers. Air traffic control wants separation assured with a workload the sector team can sustain, and a traffic picture that stays predictable. The flight crew wants instructions the aircraft can fly, given with enough time to set them up.

Some of these objectives align. Burning less fuel lowers the airline's cost and lowers carbon dioxide emissions together. Air traffic control's wish for predictability and the airline's wish to keep its schedule both favor aircraft that fly what was planned.

Others conflict, and the conflict this project studies is the first of them: the trajectory that is cheapest for one aircraft is not always the one that costs the sector least to grant. A request or a resolution that saves fuel for one flight can add instructions, monitoring, and follow-up for the controller. That cost is externalized. It does not appear in the airline's own cost function, and the party that bears it has no means to price it back.

The objectives also operate on different timescales. An airline sets its schedule policy over months, realizes fuel cost per flight, and realizes delay cost in minutes. Air traffic control manages capacity over years through sector design and over seconds through tactical control. A decision made in minutes by one party is judged against objectives the others hold on other clocks.

Plan ref: to-do-list.md S9 (owner's note, 2026-10-11 print: "Not really here yet. Can show prelim results from my table") — The myopic-optimization analysis (definition of a local optimum in the system-of-systems setting, candidate system-level measures of value, the documented conflicts) is not written. For this report, show it through the one conflict the experiment measures: read across the rows of Table 6 for an option that is better for one aircraft or airline and worse for the sector or the system. The other candidate conflicts are future work (D-016) and go to Appendix A.4.

4.3 Architecture: Detailed and Abstracted

Plan ref: to-do-list.md S10 — Present the SysML architecture: package structure, the context diagram (Figure 4), the internal block diagrams below, the interface and item-flow model, and the requirements model with traceability. For the aircraft, the detailed diagram comes first and the abstracted one second (your note: "Place detailed and abstracted"); say what the abstraction drops and why. The airspace management diagram is still a placeholder.

AI-DRAFTED, owner to rewrite — Two connecting sentences so the figures are referred to from the text. Replace with your own description of what each level of detail shows.

The aircraft is shown at two levels of detail. Figure 10 is the detailed internal block diagram, including propulsion, airframe, landing gear, utilities, and the fuel and power links. Figure 11 is the abstraction used in the rest of the report: the crew, the flight deck controls, the avionics, the flight controls, the fuel system, and the information and command links among them, which are the parts the studied decision passes through. Figure 12 shows the internals of airspace management as modeled so far.

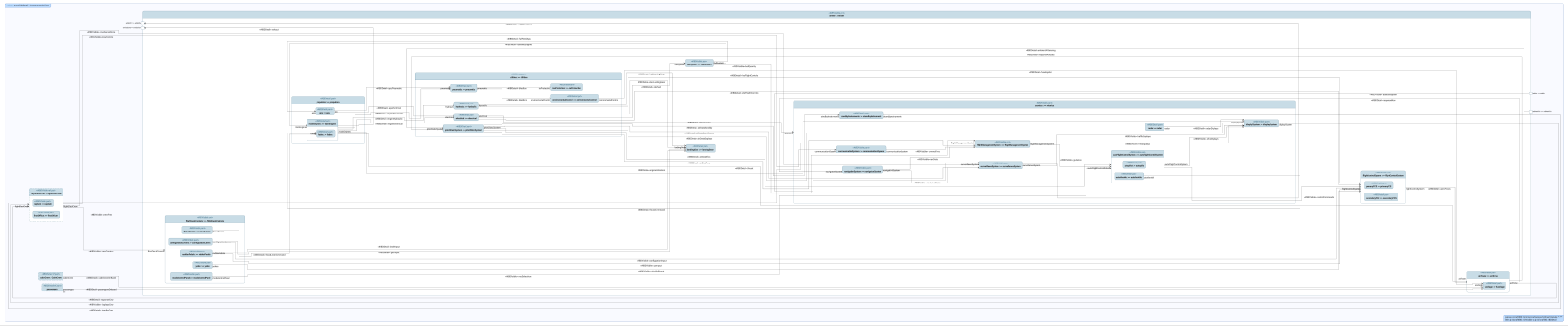


Figure 10: Aircraft internal block diagram, detailed level: adds propulsion, airframe, landing gear, utilities, and the fuel and power links. View `aircraftIbdDetail`; source `nas_ibd_aircraft.sysml`; rendered 2026-10-10.

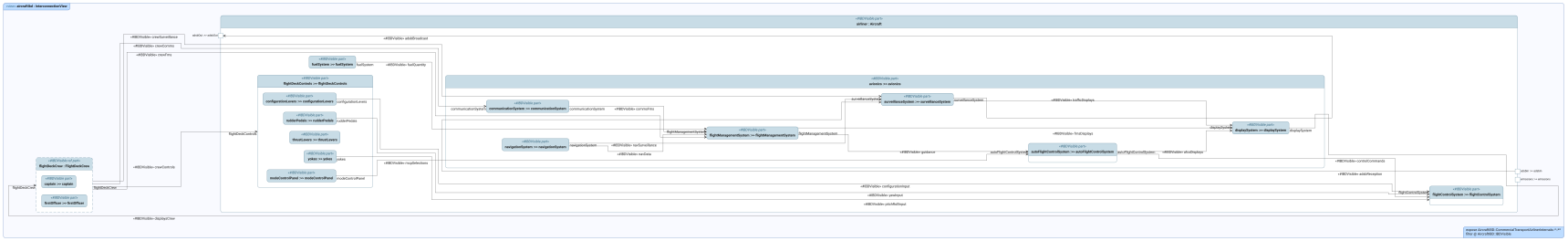


Figure 11: Aircraft internal block diagram, default level of detail. View `aircraftIbd`; source `nas_ibd_aircraft.sysml`; rendered 2026-10-10.

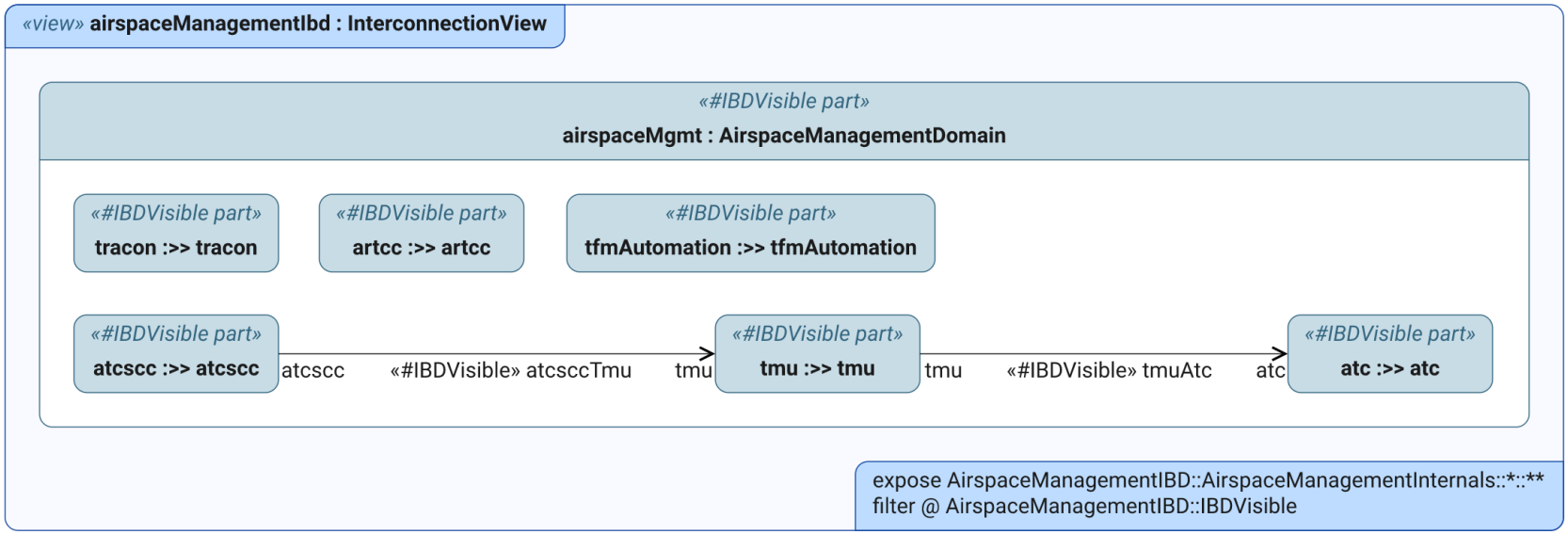


Figure 12: Airspace management internal block diagram. View `airspaceManagementIbd`; source `nas_ibd_airspace_management.sysml`; rendered 2026-10-10.

4.4 Experiment Results

Table 6: Outcome of each case at four levels, for the base scenario scored at the actual masses. Each cost is a difference from flying on undisturbed. Preliminary: the scenario's numbers are not all confirmed.

Case	Option	Largest cost to any one aircraft (kg fuel-equiv.)	Airline totals (USD)	Sector (calls / effort / risk)	System fuel, CO2 (kg)
A	O8	32 (AC1)	X 28, Y 0, Z 0	1+1 / 1 / low	32, 101
B	O9	11 (AC1)	X 15, Y 0, Z 0	2+2 / 3 / medium	16, 51
C	O4	12 (AC2)	X 0, Y 11, Z 0	1+1 / 3 / low	12, 39
D	O4	12 (AC2)	X 0, Y 11, Z 0	1+1 / 3 / low	12, 39

Plan ref: to-do-list.md S12 — Which option each case chose, and why. Walk Table 6 row by row. For case A, name the rule that produced the choice. For B, C and D, name the cost that decided it.

Plan ref: to-do-list.md S12 — Did the climb rule matter? With the crossing pair above FL350, cases A and B could not offer a climb and cases C and D could. Say whether case C used one, for which aircraft, and what it saved. If C did not climb either, say so: that is a result too.

Plan ref: to-do-list.md S12 — B minus A, and C minus B. The first is what automation is worth; the second is what the information exchange is worth. Compare the size of each with the prior figure of tens of gallons per flight ([19], p. 12) and say whether the difference is larger than the truth model's uncertainty.

Plan ref: to-do-list.md S9, S12 — Does the best option change with the level? Read across each row of Table 6. Name any option that is better for one aircraft or one airline and worse for the sector or the system, and who pays (the largest cost to any one aircraft). This is the sub-system versus system discussion.

Plan ref: to-do-list.md S12 — Airline X's two flights. Discuss the option that touches both of one airline's aircraft: what the airline would choose between its late flight and its early one, and that the controller cannot see that preference. Connect to the dispatcher priorities (schedule integrity and connections rated at or above fuel; source of that table still to confirm).

Plan ref: to-do-list.md S12 — Workload is three things. Discuss clearances issued, effort to devise, and residual risk separately. Say where a "more workload" option on one of them is "less" on another.

Plan ref: to-do-list.md S12 — When would the controller say no? (owner's note, second hand review, 5.2). Discuss whether refusals are more likely in morning rushes at hub airports or near holidays, using the sector-loading rule in [19] (p. 10). A hypothesis; needs a source.

Plan ref: to-do-list.md S12 — Is schedule the right priority? (owner's note, second hand review, 4.11). Contrast a carbon price, which makes fuel weigh more against time, with controllers' usual practice of giving schedule the priority.

Plan ref: to-do-list.md Track A — Figure slot: sequence diagram of case C's conflict resolution (controller, ATC decision support, airline operations center, FMS, crew), rendered from the model once drafted (turn 3).

Plan ref: to-do-list.md S12 — The case where nothing changes. Report the variation with similar weights and the climb closed, where case C picks the same option as case B, to show the comparison is not rigged.

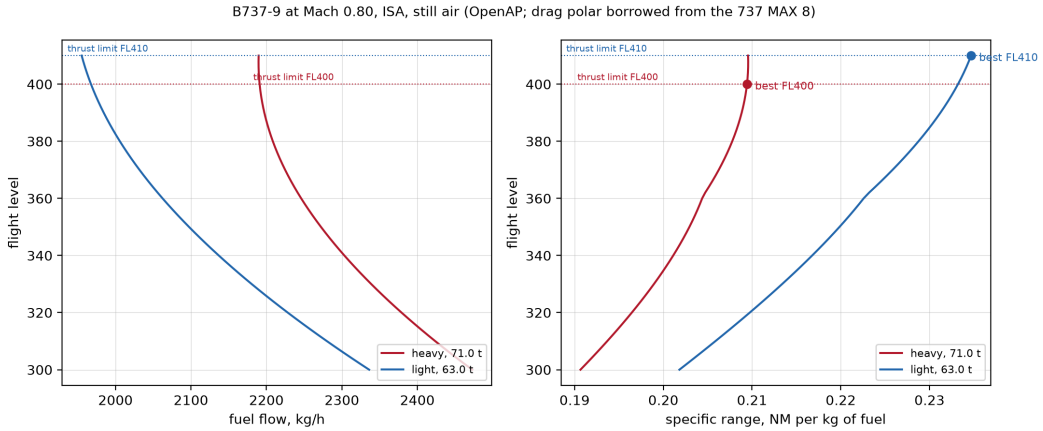


Figure 13: Fuel flow and specific range against altitude at Mach 0.80 for the heavy and light B737-9 (OpenAP, D-017).

4.5 Trace Back to the Model

Doesn't need its own section.

AI-DRAFTED, owner to rewrite — Where each column of the outcome table lives in the model (your note: "DO"). Part and port names are from `cameo_models/nas_context_diagram.sysml`; the requirement and its tag are from `knowledge/models/requirements-and-traceability.md`. Check the part chosen for each column.

Each column of Table 6 belongs to a part of the context diagram (Figure 4). The aircraft column belongs to the aircraft and its flight crew. The airline column belongs to the airline operations center within flight operations. The sector column belongs to air traffic control within airspace management. The system column belongs to no single part. It is a property of the NAS as a whole, which is the reason no constituent system computes it today and the reason the model is needed to state it.

The exchange that cases C and D add is one interface in the model: the airline operations center's performance share to air traffic control's decision support, carrying gross weight, cost index and the flight's delay margin. It satisfies system requirement SLR-INF-02, which is tagged as part of the improved architecture because today's NAS does not meet it. The interface is not drawn on Figure 4, which shows the baseline.

4.6 Emissions and Who Pays

Plan ref: to-do-list.md S9, S12 — Airline level: put CO2 into dollars. Take the fuel difference between cases from Table 6, convert to kg of CO2, then to dollars. State the route used and its source. Candidates: (1) the price of a carbon credit or allowance, if a scheme applies to the flight; (2) passenger value, meaning revenue gained or lost because some passengers choose the flight listed with lower emissions. Make the point that fuel is already counted once as fuel cost and that pricing its CO2 counts the same kilograms again, on purpose. SOURCES NOT YET HELD.

Plan ref: to-do-list.md S9, S12 — Passenger value. Discuss how passengers see a flight's emissions when booking, and whether that reaches the airline as demand. The author's own practice of choosing lower-emission listings is the motivation, not the evidence: cite a study. Note the gap between one decision in one sector and the label a passenger saw at booking; the link is the airline's average over many flights.

Plan ref: to-do-list.md S12 — Scale it honestly. One conflict in one sector is a small number. If a per-year or fleet-wide figure is offered, state the multiplier and its source (for example, how often level crossing conflicts occur), and label it illustrative.

Plan ref: to-do-list.md S9 — Who pays. For the option the system prefers, name the aircraft or airline that is worse off and by how much. This is the equity discussion.

5 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its experiment. Translate the conflict analysis and the experiment's results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, instructions issued) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to the objectives whose costs are externalized onto another stakeholder, Section 4.2, and to Section 4.6).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template's sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a "before" baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

6 Conclusions and Recommendations

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., "reduces network-level delay propagation by an illustrative X% under the demonstrated scenario" rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the experiment (Sections 2.2 and 3.4), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

Plan ref: to-do-list.md S14, S15 — Recommendations. Restate the main findings from Section 4 and provide recommendations for moving forward. Include the alternatives considered (alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

References

[1] Airbus Customer Services. *Flight Operations Briefing Notes: Approach Techniques: Aircraft Energy Management during Approach*. Tech. rep. FLT_OPS – APPR – SEQ03 – REV02. Airbus, Oct. 2005. URL: `https://skybrary.aero/sites/default/files/bookshelf/166.pdf`.

[2] Magnus Axholt and Stephen Peterson. "Modelling Traffic Scenarios for Realistic Air Traffic Control Environment Testing". MA thesis. Linköping University, 2004.

[3] Matthias Bartel and Trevor M. Young. "Simplified Thrust and Fuel Consumption Models for Modern Two-Shaft Turbofan Engines". In: *Journal of Aircraft* 45.4 (2008). Not held and not read. Authors, title, journal, volume, issue, pages and DOI confirmed 2026-10-11 against the Crossref record for the DOI (DOI supplied by the owner)., pp. 1450–1456. DOI: `10.2514/1.35589`.

[4] Jason E. Bartolomei et al. "Engineering Systems Multiple-Domain Matrix: An Organizing Framework for Modeling Large-Scale Complex Systems". In: *Systems Engineering* 15.1 (2012), pp. 41–61. DOI: 10.1002/sys.20193.

[5] Katherine A. Berry and John J. Pace. "Examining the Actors and Functions of an Airline Operations Center". In: *Proceedings of the Human Factors and Ergonomics Society 55th Annual Meeting*. Vol. 55. 1. 2011, pp. 1412–1416. DOI: 10.1177/1071181311551294.

[6] Karl D. Bilimoria, Mike M. Paglione, and Hilda Q. Lee. "Performance Analysis of a Conflict Probe Utilizing Only State Vector Information". In: *24th International Congress of the Aeronautical Sciences (ICAS)*. Yokohama, Japan, 2004. URL: https://www.icas.org/icas_archive/ICAS2004/PAPERS/216.PDF.

[7] Katie Bolczak et al. *FAA Data Standards Initiative: Systems Engineering Base for Air Traffic Modernization*. Tech. rep. The MITRE Corporation, 2001.

[8] Soufiane Bouarfa, Jasper Müller, and Henk A. P. Blom. "Evaluation of a Multi-Agent System Approach to Airline Disruption Management". In: *Journal of Air Transport Management* 71 (2018), pp. 108–118. DOI: 10.1016/j.jairtraman.2018.05.009.

[9] Chris Brady. *Rules of Thumb*. The Boeing 737 Technical Site (web page). 2026. URL: http://www.b737.org.uk/rulesofthumb.htm.

[10] Ian Britchford. "737 MAX 8 & 9 fuel burn and operating performance". In: *Aircraft Commerce* 137 (Aug. 2021), pp. 18–26. URL: https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/General%20Articles/2021/137_FLTOPS.pdf.

[11] António Jesus Monteiro de Castro. "A Distributed Approach to Integrated and Dynamic Disruption Management in Airline Operations Control". PhD thesis. Porto, Portugal: Faculdade de Engenharia, Universidade do Porto (FEUP), 2013.

[12] Mohammad Chami et al. "Towards Solving MBSE Adoption Challenges: The D3 MBSE Adoption Toolbox". In: *28th Annual INCOSE International Symposium*. Washington, DC, 2018.

[13] Gano Chatterji. *Introduction to Air Traffic Management*. Video and transcript, Aviation Data Science Seminar (virtual). NASA Technical Reports Server (NTRS), Document ID 20205001251. NASA Ames Research Center; Universities Space Research Association, Apr. 2020. URL: https://ntrs.nasa.gov/citations/20205001251 (visited on 09/20/2026).

[14] Michael D. D. Clarke. "Irregular Airline Operations: A Review of the State-of-the-Practice in Airline Operations Control Centers". In: *Journal of Air Transport Management* 4 (1998), pp. 67–76.

[15] Richard A. Coppenbarger. "En Route Climb Trajectory Prediction Enhancement Using Airline Flight-Planning Information". In: *AIAA Guidance, Navigation, and Control Conference and Exhibit*. AIAA-99-4147. 1999, pp. 1077–1087. DOI: 10.2514/6.1999-4147.

[16] Daniel A. DeLaurentis. "Understanding Transportation as a System-of-Systems Design Problem". In: *43rd AIAA Aerospace Sciences Meeting and Exhibit*. Reno, Nevada, 2005. DOI: 10.2514/6.2005-123.

[17] Daniel A. Delaurentis and Donald N. Fry. "Understanding the Implications for Airports of Distributed Air Transportation Using a System-of-Systems Approach". In: *Transportation Planning and Technology* 31.1 (2008), pp. 69–92. DOI: 10.1080/03081060701835738.

[18] Ahmed E. E. Eltoukhy, Felix T. S. Chan, and S. H. Chung. "Airline Schedule Planning: A Review and Future Directions". In: *Industrial Management & Data Systems* 117.6 (2017), pp. 1201–1243. DOI: 10.1108/IMDS-09-2016-0358.

[19] Engility Corporation. *Traffic Aware Strategic Aircrew Requests (TASAR): Annualized TASAR Benefits for Alaska Airlines Operations*. Contractor report NNL12AA06C, deliverable 41A. Corporate author, as the document states: "prepared by Engility Corporation ... under Contract No. NNL12AA06C with NASA Langley Research Center" (Preface, p. 2); NASA Technical Monitor: David Wing (title page). No personal author is named. A web search summary of the NASA record for a later published version (NASA/CR-2015-218787, NTRS 20150017048) names Jeffrey Henderson; not confirmed and not used here. Engility Corporation, Billerica, MA, for NASA Langley Research Center, Sept. 2014. URL: https://ntrs.nasa.gov/api/citations/20140012787/downloads/20140012787.pdf.

[20] EUROCONTROL. *EUROCONTROL Specification for Airport Collaborative Decision Making (A-CDM)*. Tech. rep. Brussels, Belgium: EUROCONTROL, 2025.

[21] EUROCONTROL. *Aircraft Performance Database: BOEING 737 MAX 9 (B39M)*. Web page. 2026. URL: https://learningzone.eurocontrol.int/ilp/customs/ATCPFDB/details.aspx?ICAO=B39M.

[22] Federal Aviation Administration. *Aircraft Operations at Altitudes Above 25,000 Feet Mean Sea Level or Mach Numbers Greater Than .75*. Advisory Circular AC 61-107B, Change 1. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Sept. 2015. URL: https://www.faa.gov/regulations_policies/advisory_circulars.

[23] Federal Aviation Administration. *Instrument Procedures Handbook, Chapter 3: Arrivals*. Tech. rep. FAA-H-8083-16B. U.S. Department of Transportation, Federal Aviation Administration, 2017. URL: https://www.faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/instrument_procedures_handbook/FAA-H-8083-16B_Chapter_3.pdf.

[24] Federal Aviation Administration. *Flight Information Exchange Model (FIXM) US Extension v4.4.0: Logical Model Diagrams*. Tech. rep. Federal Aviation Administration / FIXM Program, 2024.

[25] Federal Aviation Administration. *Aeronautical Information Manual: Official Guide to Basic Flight Information and ATC Procedures*. Aeronautical Information Manual. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Feb. 2025. URL: https://www.faa.gov/air_traffic/publications.

[26] Federal Aviation Administration. *Air Traffic Control*. Order JO 7110.65BB. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[27] Federal Aviation Administration. *FAA Services Hierarchy*. One-page slide (PDF), FAA NAS Enterprise Architecture. Year is the PDF creation date (2025-02-24, PowerPoint export; PDF author field "ANG-B2"); no version, title-page date or byline appears on the slide. Unverified: what ANG-B2 stands for, and whether this is the current issue of the hierarchy on the FAA NAS EA site (faa.gov/about/office_org/headquarters_offices/ang/r2r) – check before citing as current. 2025.

[28] Federal Aviation Administration. *Facility Operation and Administration*. Order JO 7210.3EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[29] Federal Aviation Administration. *Flight Services*. Order JO 7110.10EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[30] Federal Aviation Administration. *NAS Enterprise Architecture: Infrastructure Roadmaps*. Tech. rep. Federal Aviation Administration, 2025.

[31] Federal Aviation Administration. *Airport Compliance Manual, Chapter 12: Review of Aeronautical Lease Agreements*. Order 5190.6C. URL not recorded when the chapter was downloaded on 2026-09-28; add it from the FAA Order 5190.6C page before citing. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026.

[32] Federal Aviation Administration. *Airport Compliance Manual, Chapter 9: Unjust Discrimination between Aeronautical Users*. Order 5190.6C. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026. URL: https://www.faa.gov/airports/resources/publications/orders/compliance_5190_6/order-5190-6c-compliance-chapter-09 (visited on 09/28/2026).

[33] Michael Finke et al. *D2.1: Current TBO Concepts and Derivation of the Green Air Traffic Management Concepts*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2020.

[34] Selina Fothergill and Andrew Neal. "Conflict-Resolution Heuristics for En Route Air Traffic Management". In: *Proceedings of the Human Factors and Ergonomics Society 57th Annual Meeting*. 2013, pp. 71–75. DOI: 10.1177/1541931213571018.

[35] GE Aerospace. *GE Aerospace Joins MBSE Ecosystem in Michigan Aerospace*. University of Michigan College of Engineering, Aerospace Engineering news. May 2023. URL: https://aero.engin.umich.edu/2023/05/11/ge-aerospace-joins-mbse-ecosystem-in-michigan-aerospace/ (visited on 09/14/2026).

[36] L. K. Hassan, B. F. Santos, and J. Vink. "Airline Disruption Management: A Literature Review and Practical Challenges". In: *Computers & Operations Research* 127 (2021), p. 105137. DOI: 10.1016/j.cor.2020.105137.

[37] Terry R. Hill et al. "NASA's Use of MBSE and SysML Modeling to Architect the Future of Human Exploration". In: *34th Annual INCOSE International Symposium*. Dublin, Ireland, 2024.

[38] Eric C. Honour. "Systems Engineering Return on Investment". In: *INCOSE International Symposium*. Vol. 20. 1. Chicago, IL, 2010, pp. 1422–1439. DOI: 10.1002/j.2334-5837.2010.tb01150.x.

[39] Eric C. Honour. "Sizing Systems Engineering Activities to Optimize Return on Investment". In: *INCOSE International Symposium*. Vol. 21. 1. Denver, CO, 2011, pp. 1233–1245. DOI: 10.1002/j.2334-5837.2011.tb01281.x.

[40] Eric C. Honour. "Systems Engineering Return on Investment". PhD thesis. Adelaide, Australia: Defence, Systems Institute, School of Electrical, and Information Engineering, University of South Australia, 2013. URL: https://segoldmine.ppi-int.com/node/72248 (visited on 09/19/2026).

[41] Niamat Ullah Ibne Hossain et al. "Modeling and Analysis of Unmanned Aerial Vehicle System Leveraging Systems Modeling Language (SysML)". In: *Systems* 10.6 (2022), p. 264. DOI: 10.3390/systems10060264.

[42] Haoliang Hu et al. *D2.2: Report about Modelling of Operational and System Architecture*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[43] Haoliang Hu et al. *D5.1: Report on ATM Avionic System Architecture Development*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[44] Yuzhen Hu et al. "Review of Optimization Problems, Models and Methods for Airline Disruption Management from 2010 to 2024". In: *Digital Transportation and Safety* 3.4 (2024), pp. 246–263. DOI: 10.48130/dts-0024-0022.

[45] INCOSE. *Guide to Writing Requirements*. Tech. rep. INCOSE-TP-2010-006-04. Version 4. International Council on Systems Engineering, 2023.

[46] INCOSE. *INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*. 5th. Wiley, 2023. ISBN: 978-1-119-81429-0. URL: https://www.incose.org/about-systems-engineering/system-and-se-definition/systems-engineering-definition (visited on 09/19/2026).

[47] International Civil Aviation Organization. *Global Air Traffic Management Operational Concept*. Doc 9854, AN/458. Montreal, Canada: International Civil Aviation Organization, 2005.

[48] International Civil Aviation Organization. *ICAO Carbon Emissions Calculator Methodology*. Methodology document Version 13.1. International Civil Aviation Organization, Aug. 2024. URL: https://www.icao.int/sites/default/files/environmental-protection/ENVtools/SuportingMaterials/ICEC_Methodology_Passagers_v13_Final.pdf.

[49] International Civil Aviation Organization. *ICAO Aircraft Engine Emissions Databank*. Spreadsheet, hosted by the European Union Aviation Safety Agency. Issue 32. Mar. 2026. URL: https://www.easa.europa.eu/easa-and-you/environment/icao-aircraft-engine-emissions-databank.

[50] ISO/IEC/IEEE. *ISO/IEC/IEEE 15288:2023 Systems and software engineering — System life cycle processes*. Tech. rep. Geneva: International Organization for Standardization, 2023.

[51] Swapnil Sarjerao Jagtap et al. "Model-Based Systems Engineering Approach for a Systematic Design of Aircraft Engine Inlet". In: *AIAA SciTech Forum 2025*. 2025. DOI: 10.2514/6.2025-1410.

[52] Priti Jain. "Personal Knowledge Management: The Foundation of Organisational Knowledge Management". In: *South African Journal of Libraries and Information Science* 77.1 (2011), pp. 1–14. DOI: 10.7553/77-1-62.

[53] Abdessamad Jarrar and Youssef Balouki. "Formal Modeling of a Complex Adaptive Air Traffic Control System". In: *Complex Adaptive Systems Modeling* 6.1 (2018), p. 6. DOI: 10.1186/s40294-018-0056-4.

[54] Heiko Kern, Axel Hummel, and Stefan Kühne. "Towards a Comparative Analysis of Meta-Metamodels". In: *Proceedings of the Compilation of the Co-located Workshops on DSM'11, TMC'11, AGERE! 2011, AOOPES'11, NEAT'11, & VMIL'11 (SPLASH 2011 Workshops)*. ACM, 2011, pp. 7–12. DOI: 10.1145/2095050.2095053.

[55] Barry Kirwan and Mary Flynn. "Identification of Air Traffic Controller Conflict Resolution Strategies for the CORA (Conflict Resolution Assistant) Project". In: *4th USA/Europe Air Traffic Management R&D Seminar*. Santa Fe, NM, Dec. 2001.

[56] Konstantia Kontodimou et al. "Proactive Aircraft Turnaround Buffer Optimization Integrating Machine Learning and Scenario Analysis". In: *Journal of the Air Transport Research Society* 6 (2026), p. 100117. DOI: 10.1016/j.jatrs.2026.100117.

[57] Deepak Kulkarni. *Impact of Slot Substitution on CTOP Performance*. Technical Memorandum NASA/TM-2019-220158. Moffett Field, CA: NASA Ames Research Center, Mar. 2019.

[58] Tianning Liu et al. "Model-Based Systems Engineering Supporting Architecture Modeling of Air Traffic Management System and Model Verifying Based on SMT". In: *Knowledge Science, Engineering and Management (KSEM 2024)*. Vol. 2269. Communications in Computer and Information Science. Springer, 2025, pp. 183–197. DOI: 10.1007/978-981-96-0178-3_13.

[59] Manoj Lohatepanont and Cynthia Barnhart. "Airline Schedule Planning: Integrated Models and Algorithms for Schedule Design and Fleet Assignment". In: *Transportation Science* 38.1 (2004), pp. 19–32. DOI: 10.1287/trsc.1030.0026.

[60] Jing Lu et al. "Harnessing Digital Twin Technology for Enhanced Aircraft Turnaround Efficiency". In: *Transport Economics and Management* 3 (2025), pp. 334–345. DOI: 10.1016/j.team.2025.09.001.

[61] Jinzhi Lu et al. "Design Ontology Supporting Model-based Systems-engineering Formalisms". In: *arXiv preprint* (2020).

[62] Jinzhi Lu et al. "Semantic Model-based Systems Engineering based on KARMA: A Research and Practice Roadmap 2025". In: *32nd Annual INCOSE International Symposium*. Detroit, MI, 2022.

[63] Christopher A. Lupp et al. "A Theoretical Foundation and Practical Demonstration of Integrating MDO, MBSE and the Digital Thread". In: *AIAA SciTech Forum*. Air Force Research Laboratory / University of Dayton Research Institute. 2026.

[64] Apoorv Maheshwari, C. Robert Kenley, and Daniel A. DeLaurentis. "Creating Executable Agent-Based Models Using SysML". In: *25th Annual INCOSE International Symposium (IS2015)*. Bellevue, WA, 2015.

[65] N. A. M. Maiden et al. "Model-Driven Requirements Engineering: Synchronising Models in an Air Traffic Management Case Study". In: *16th International Conference on Advanced Information Systems Engineering (CAiSE 2004)*. Vol. 3084. Lecture Notes in Computer Science. Riga, Latvia: Springer, 2004, pp. 368–383.

[66] Tom McDermott et al. "AI4SE and SE4AI: A Research Roadmap". In: *INSIGHT* 23.1 (2020), pp. 8–14. DOI: 10.1002/inst.12278.

[67] Faida Mhenni et al. "Flight Control System Modeling with SysML to Support Validation, Qualification and Certification". In: *13th IFAC and IEEE Conference on Programmable Devices and Embedded Systems*. Vol. 49-3. IFAC-PapersOnLine. Elsevier, 2016, pp. 453–458. DOI: 10.1016/j.ifacol.2016.07.076.

[68] Yaniv Mordecai. "Conceptual Modeling of Cyber-Physical Gaps in Air Traffic Control". In: *Complex Adaptive Systems Conference with Theme: Cyber Physical Systems and Deep Learning (CAS 2018)*. Vol. 140. Procedia Computer Science. Elsevier, 2018, pp. 21–28. DOI: 10.1016/j.procs.2018.10.288.

[69] Ryota Mori. "Aircraft Mass Estimation Using Cruise Flight Profile". In: *33rd Congress of the International Council of the Aeronautical Sciences (ICAS)*. Stockholm, Sweden, 2022. URL: https://www.icas.org/icas_archive/ICAS2022/data/papers/ICAS2022_0105_paper.pdf.

[70] Sai V. Mudumba et al. "Model-Based Systems Engineering Approach for Simulating Urban Air Mobility Maturity Level 5 (UML-5) Operations". In: *AIAA AVIATION Forum*. 2022. DOI: 10.2514/6.2022-4075.

[71] Pamela Munro and Richard Mogford. "Managing Variability: A Cognitive Ethnography of the Work of Airline Dispatchers". In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. Vol. 62. 1. 2018, pp. 182–186. DOI: 10.1177/1541931218621043.

[72] Nadine Pilon, Laurent Guichard, and Katherine Cliff. "Reducing Impact of Delays using Airspace User-Driven Flight Prioritisation: User Driven Prioritisation Process Validation Simulation and Results". In: *9th SESAR Innovation Days*. Paper 39. ISSN 0770-1268. Dec. 2019. URL: https://sesarju.eu/sites/default/files/documents/sid/2019/papers/SIDs_2019_paper_39-1.pdf.

[73] Esa M. Rantanen and Christopher D. Wickens. "Conflict Resolution Maneuvers in Air Traffic Control: Investigation of Operational Data". In: *The International Journal of Aviation Psychology* 22.3 (2012), pp. 266–281. DOI: 10.1080/10508414.2012.691048.

[74] Bill Roberson. "Fuel Conservation Strategies: Cost Index Explained". In: *Boeing AERO* QTR_02 (2007), pp. 26–28. URL: https://skybrary.aero/sites/default/files/bookshelf/1956.pdf.

[75] Ítalo Romani de Oliveira et al. "A Predictive Services Architecture for Efficient Airspace Operations". In: *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. IEEE. 2026. DOI: 10.1109/ICNS69853.2026.11570268.

[76] Ahmed R. Sadik et al. "Urban Air Mobility as a System of Systems: An LLM-Enhanced Holonic Approach". In: *20th Annual System of Systems Engineering Conference (SOSE)*. IEEE. 2025.

[77] Mateus Santana et al. "The Aircraft Recovery Problem: A Systematic Literature Review". In: *EURO Journal on Transportation and Logistics* 12 (2023), p. 100117. DOI: 10.1016/j.ejtl.2023.100117.

[78] Michael Schmidt. "A Review of Aircraft Turnaround Operations and Simulations". In: *Progress in Aerospace Sciences* 92 (2017), pp. 25–38. DOI: 10.1016/j.paerosci.2017.05.002.

[79] Charles A. Schultz, David Thipphavong, and Heinz Erzberger. "Adaptive Trajectory Prediction Algorithm for Climbing Flights". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA-2012-4931. Venue and paper number unconfirmed from the PDF; DOI not found. 2012. URL: https://ntrs.nasa.gov/api/citations/20140004896/downloads/20140004896.pdf.

[80] Florian Schummer and Maximillian Hyba. "An Approach for System Analysis with Model-based Systems Engineering and Graph Data Engineering". In: *Data-Centric Engineering* 3 (2022), e33. DOI: 10.1017/dce.2022.33.

[81] Thomas L. Seamster, Eric Chevalley, and Barbara G. Kanki. *Collaborative Systems Assessment: Flightdeck, Air Traffic Control, Flight Operations Center and Automation*. Draft Final Report, FAA task 05-04 / AJP61SSP-0050. NASA Ames Research Center, Aug. 2011.

[82] SESAR Joint Undertaking. *SESAR Solutions Catalogue 2019*. Tech. rep. Catalogue number MG-07-18-083-EN-N. Luxembourg: Publications Office of the European Union, 2019. DOI: 10.2829/998701.

[83] SESAR Joint Undertaking. *European ATM Master Plan: Making Europe the Most Efficient and Environmentally Friendly Sky to Fly in the World*. Tech. rep. SESAR Joint Undertaking, 2025.

[84] SESAR Joint Undertaking. *SESAR eATM ATM Capability Model (Extracted Capability Catalogue)*. 2025.

[85] Kapil Sheth and Sebastian Gutierrez-Nolasco. "Analysis of Factors for Incorporating User Preferences in Air Traffic Management: A System Perspective". In: *27th International Congress of the Aeronautical Sciences (ICAS 2010)*. Nice, France, 2010.

[86] Kapil S. Sheth et al. "Simulations of Credits Concept with User Input for Collaborative Air Traffic Management". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA 2010-8079. Toronto, Ontario, Canada, Aug. 2010. DOI: 10.2514/6.2010-8079.

[87] Sonali Sinha Roy et al. "From Ontology to System Architecture: An MBSE Approach Toward the Realization of Urban Air Mobility". In: *34th Congress of the International Council of the Aeronautical Sciences (ICAS)*. Florence, Italy, 2024.

[88] Junzi Sun, Jacco M. Hoekstra, and Joost Ellerbroek. "OpenAP: An Open-Source Aircraft Performance Model for Air Transportation Studies and Simulations". In: *Aerospace* 7.8 (2020), p. 104. DOI: 10.3390/aerospace7080104. URL: https://openap.dev (visited on 10/10/2026).

[89] Serkan Turhan, Fatma Gzara, and Samir Elhedhli. "Workload Balancing for Flight Dispatchers". In: *Computers & Operations Research* 186 (2026), p. 107303. DOI: 10.1016/j.cor.2025.107303.

[90] Sabine Wilke, Arnab Majumdar, and Washington Y. Ochieng. "Airport Surface Operations: A Holistic Framework for Operations Modeling and Risk Management". In: *Safety Science* 63 (2014), pp. 18–33. DOI: 10.1016/j.ssci.2013.10.015.

[91] Tim Wittenborg et al. *Knowledge-Based Aerospace Engineering: A Systematic Literature Review*. 2025.

[92] Yifan Xu, Sebastian Wandelt, and Xiaoqian Sun. "Airline Scheduling Optimization: Literature Review and a Discussion of Modelling Methodologies". In: *Intelligent Transportation Infrastructure* 2 (2024), liad026. DOI: 10.1093/itinfr/liad026.

[93] Anzhuo Yao et al. "System-of-Systems Safety for Low-Altitude Aviation Transportation". In: *Reliability Engineering and System Safety* 273 (2026), p. 112276. DOI: 10.1016/j.ress.2026.112276.

[94] Haytham Younus et al. "AI- and Ontology-Based Enhancements to FMEA for Advanced Systems Engineering: Current Developments and Future Directions". In: *Applied Sciences* 16.5 (2026), p. 2464. DOI: 10.3390/app16052464.

A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 4.1).

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in the body: full BDD/IBD set, requirements model, and complete traceability matrices (needs → objectives → requirements → systems → activities → information exchanges → aircraft behavior). Not yet modeled as views: BDDs and the requirements model.

Every figure of the model in this report is a view defined in `nas_package_my_views.sysml` and rendered from the SysML v2 text with Syside. Its caption names the view, the `.sysml` file that holds the content, and the date of the render. As of this draft all five views appear in the body: Figure 4 (Section 1.4), Figure 9 (Section 4.1), and Figures 10, 11 and 12 (Section 4.3). Views added to the model later appear below.

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 4.

Plan ref: to-do-list.md S11 — Full options table: for each of the nine options, the maneuver per aircraft, instructions now, follow-up later, and what the decider must know to use it. Draft in `knowledge/models/experiment-options-and-scoring.md` §2.

Plan ref: to-do-list.md S11, S12 — Every option scored at four levels (the table from which each case picks one row), the per-viewpoint cost statements with their symbols defined, and the table of abstractions with reason and risk.

A.6 Generative AI Use Disclosure

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Generative AI disclosure: owner to draft. Required by the course policy whenever GenAI is used; Interim Report #2 is the first submission after the policy was read (2026-09-21).

A.7 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (`evidence/source-register.md`): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.